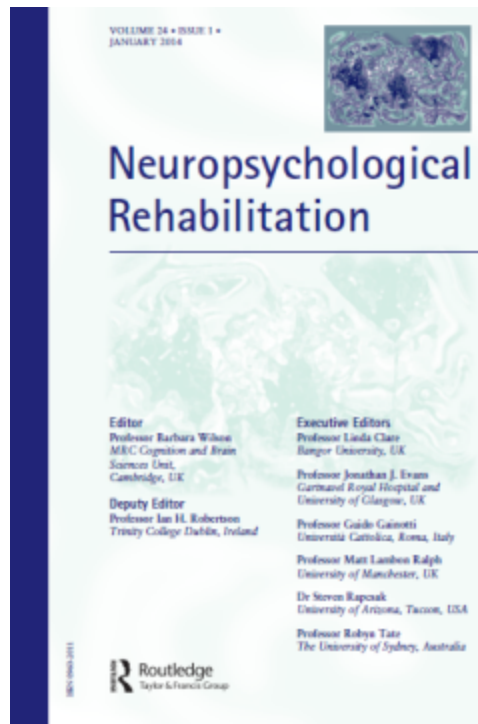




Kinesthesia: feasibility and preliminary efficacy of a sensory-motor cognitive intervention in community-dwelling older adults with and without Mild Cognitive Impairment

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04 July 2026

The Editor-in-Chief

Neuropsychological Rehabilitation

Dear Editor-in-Chief,

We are pleased to submit our manuscript, “Kinesthesia: feasibility and preliminary efficacy of a sensory-motor cognitive intervention in community-dwelling older adults with and without Mild Cognitive Impairment,” for consideration as an original empirical article in Neuropsychological Rehabilitation.

Kinesthesia is a low-cost therapeutic-videogame platform that couples cognitive training with whole-body kinematic interaction, designed for older adults across the cognitively healthy–MCI spectrum. In this single-arm feasibility study, 20 participants (11 cognitively healthy, 9 with MCI) completed a 30-session programme (60 minutes, three times weekly, over ten weeks) with 100% adherence. Using a continuous cognitive-reserve framework (MoCA as a covariate) and permutation-based inference with Benjamini–Hochberg correction, we observed BH-corrected improvements across verbal-memory and executive NEUROPSI indices (Cohen’s d up to ~ 1.1) and in both six-minute-walk outcomes (combined $d = 1.20$). The normalised global cognitive score improved into the high-normal range specifically in the cognitively healthy subgroup ($d = 1.38$), whereas gains in the MCI subgroup concentrated in verbal memory and executive sequencing — a reserve-dependent pattern consistent with the study’s central hypothesis. We interpret these as preliminary, hypothesis-generating findings; the absence of a randomised control condition is discussed explicitly as the principal limitation.

We believe the work is well suited to Neuropsychological Rehabilitation, which publishes research on rehabilitation, recovery of function, cognitive training, and brain plasticity. Kinesthesia is precisely a scalable cognitive-motor rehabilitation tool, and our findings speak to reserve-dependent mechanisms of cognitive change in early decline—a question of direct relevance to the journal’s readership.

In the interest of full transparency, a brief proof-of-concept summary of this cohort appeared as a Letter to the Editor in Archives of Gerontology and Geriatrics (doi:10.1016/j.archger.2026.106346). The present manuscript is the full companion report: it contains the complete methodology, the per-domain results, the cognitive-reserve analyses, and the supplementary materials not included in the Letter, and it does not reproduce any figure or table from it. A companion paper reporting the resting-state EEG correlates of the same cohort is under separate consideration and is cited accordingly.

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This manuscript is original, is not under consideration for publication elsewhere, and all named authors have read and approved its submission. The study was approved by the Research Ethics Committee of the Universidad Autónoma de Querétaro, and all participants provided written informed consent. The authors declare no competing interests.

We would be grateful for your consideration and would be happy to suggest potential reviewers on request.

Yours sincerely,

Jorge F. Bosch-Bayard, PhD
On behalf of all co-authors

For Peer Review Only

Kinesthesia: feasibility and preliminary efficacy of a sensory-motor cognitive intervention in community-dwelling older adults with and without Mild Cognitive Impairment

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Abstract

Mild Cognitive Impairment (MCI) represents a critical window for non-pharmacological intervention, yet scalable, low-cost platforms combining cognitive and sensory-motor training remain scarce. We evaluated the feasibility and preliminary efficacy of Kinesthesia — a neuroplasticity-oriented platform using adaptive therapeutic video games with full-body interaction — in 20 community-dwelling older adults spanning the cognitively healthy (n=11) to MCI (n=9) spectrum (mean age 69.7±3.6 years; 86.4% women). In this single-arm pre-post pilot, participants completed 30 sessions (60 minutes, three times weekly, over ten weeks). Cognitive and functional outcomes were assessed with the NEUROPSI Attention and Memory battery and the Six-Minute Walk Test; pre-to-post change was evaluated by sign-flip permutation testing with Benjamini-Hochberg correction, using MoCA as a continuous cognitive-reserve covariate. Adherence was 100%. BH-corrected improvements emerged across verbal-memory and executive indices (Cohen's d up to ~1.1) and in both walking outcomes (combined d=1.20). The normalised global score improved into the high-normal range specifically in the cognitively healthy subgroup (d=1.38), whereas MCI gains concentrated in verbal memory — a reserve-dependent pattern.

These preliminary, hypothesis-generating findings support a confirmatory randomised controlled trial.

Keywords: mild cognitive impairment; cognitive rehabilitation; cognitive-motor training; exergaming; neuroplasticity; feasibility study

1. Introduction

The gap between life expectancy and disability-free years — approximately 10 years in high-income countries and proportionally larger in low- and middle-income settings — is driven primarily by cognitive decline, including Mild Cognitive Impairment (MCI) and dementia (Livingston et al., 2020; World Health Organization, 2019). MCI represents a critical preclinical window: approximately 50–70% of affected individuals progress to dementia within 5–7 years, though reversion to normal cognition is also documented, making early intervention both feasible and potentially high-impact (Menéndez González, 2014; Mitchell & Shiri-Feshki, 2009). In the absence of curative pharmacological treatments, international consensus supports multicomponent, non-pharmacological strategies combining physical activity, cognitive stimulation, and social engagement (Alzheimer’s Association, 2023; Livingston et al., 2020).

A central insight from contemporary neuroscience is that cognitive and motor functions are deeply interdependent, sharing overlapping neural substrates across frontoparietal, basal ganglia, thalamic, and cerebellar systems (Doyon & Benali, 2005; Leisman et al., 2016). This interdependence provides the mechanistic rationale for simultaneous sensory-motor cognitive interventions: co-activation of shared networks may induce broader neuroplastic changes than either cognitive or physical training alone. Meta-analyses of combined cognitive–physical programmes confirm improvements in executive function, balance, and gait in MCI, with simultaneous protocols generally outperforming sequential approaches (Karssemeijer et al., 2017; Wollesen & Voelcker-Rehage, 2014). Exergame-based interventions similarly show positive effects on cognition and functional mobility (Gallou-Guyot et al., 2020). However, effect sizes in controlled trials remain modest ($g=0.21–0.37$ for global cognition), and evidence is heterogeneous regarding adherence, real-world transfer, and optimal dosing (Gallou-Guyot et al., 2020; Karssemeijer et al., 2017; Wollesen & Voelcker-Rehage, 2014). Existing platforms often increase implementation costs without consistently superior outcomes, and few have been designed from the outset for scalable deployment in community and low-resource settings.

Neurophysiological evidence further supports the mechanistic plausibility of such approaches: EEG studies show that sensory-motor cognitive dual-task protocols produce measurable changes in movement-related cortical potentials and alpha/beta event-related desynchronisation alongside improvements in inhibitory control (Chiarenza, 1982; Chiarenza & Bosch-Bayard, 2025; Ali et al., 2022). Quantitative EEG indices have demonstrated clinical utility as biomarkers along the

healthy-to-MCI-to-dementia continuum (Babiloni et al., 2009; Smailovic & Jelic, 2019), offering a neurophysiological lens through which intervention-induced plasticity can be characterised.

In this context, we developed Kinesthesia — a sensory-motor cognitive platform based on adaptive therapeutic video games and full-body kinematic interaction — designed to simultaneously engage executive, attentional, mnemonic, and motor systems within each task. Kinesthesia runs on commercially available hardware with affordable motion-capture sensors, targeting deployment across clinical, community, and home-based settings in both high- and low-resource contexts. This paper reports feasibility and preliminary pre-to-post efficacy data in 20 community-dwelling older adults spanning the healthy to MCI spectrum, including brief resting-state EEG findings recorded in the same sample. A brief proof-of-concept communication summarising primary outcomes from this dataset has been accepted for publication as a Letter to the Editor (Bosch-Bayard et al., 2026); the present manuscript constitutes the complete scientific report with comprehensive methodology, full statistical characterisation, and extended theoretical analysis.

2. Methods

2.1 Study design

We conducted a single-arm, pre-post pilot study with repeated measures. The primary analysis treated participants as a single continuous sample ($N=20$), using the Montreal Cognitive Assessment (MoCA) score as a continuous index of cognitive reserve. Exploratory subgroup comparisons (HC: $\text{MoCA} \geq 26$; MCI: $\text{MoCA} < 26$) are reported as secondary evidence. All procedures complied with the Declaration of Helsinki; ethics approval was granted by the Ethics Committee of the Faculty of Medicine, Universidad Autónoma de Querétaro (protocols 02-03/11/2016 and 03-01/12/2016; institutional grant FOPER No. 143834).

2.2 Participants

Seventy older adults attending the Plan Vida community day centre in Querétaro, Mexico, were screened in February 2017. Inclusion criteria: age ≥ 60 years; independent ambulation without assistive devices; no uncontrolled diabetes mellitus or arterial hypertension; no active cardiopulmonary disease; absence of neurodegenerative disease, thyroid disorder, or anaemia; absence of clinically significant depression (Yesavage Geriatric Depression Scale < 5); no hazardous alcohol use (AUDIT < 8); no moderate-to-severe uncorrected sensory impairment; and attainment of a minimum 6MWT threshold for safety. Of 70 screened, 22 met all criteria and enrolled; of these, 20 had complete pre- and post-intervention outcome data ($n=11$ HC, $n=9$ MCI) and constitute the analysis sample. Two MCI participants did not complete post-intervention assessment for non-intervention-related reasons.

The sample comprised 86.4% women, mean age 69.7 ± 3.6 years (range 62–76), MoCA range 14–29 (mean 24.8 ± 3.7). HC participants had a mean age of 70.1 ± 3.8 years and 13.9 ± 2.5 years of education; MCI participants had a mean age of 69.3 ± 3.4 years and 7.6 ± 3.0 years of education. The subgroups differed substantially in education ($t=4.92$, $p<0.001$). Spearman correlation between

MoCA and education was $p=0.50$ ($p=0.024$), driven predominantly by between-group differences. MoCA was retained as the sole cognitive reserve covariate to avoid multicollinearity ($N=20$); educational stratification is planned for the confirmatory RCT.

Cognitive status was established using the MMSE for initial eligibility screening and the MoCA (Spanish version; MCI threshold <26) for subgroup classification. Neither instrument was used as a repeated-measures outcome variable, given well-documented ceiling effects and limited sensitivity to short-term change (Franco-Marina et al., 2010; Malek-Ahmadi et al., 2019). MCI classification based solely on MoCA, without full neuropsychological battery confirmation or neuroimaging, does not fully meet current NIA-AA or DSM-5 criteria; this is acknowledged as a limitation.

2.3 The Kinesthesia platform

Kinesthesia is a sensory-motor cognitive platform designed to operationalise six established neuroplasticity principles simultaneously within each session — including intensity of engagement, specificity of challenge, adaptive difficulty, salience-driven motivation, and transfer-appropriate processing (Nahum et al., 2013) — through therapeutic video games in which cognitive and motor demands are structurally inseparable: participants simultaneously navigate space, maintain dynamic postural balance, coordinate upper and lower limbs, and perform cognitive operations throughout each task. Unlike sequential dual-task protocols or post-hoc adapted exergames, Kinesthesia requires concurrent cognitive and motor processing continuously, without alternating periods of rest from either domain. A full description of the game suite (five functional categories), the adaptive difficulty algorithm, and the session structure is provided in Supplementary Materials S3.

2.4 Proposed neurobiological mechanisms

At the network level, simultaneous sensorimotor and cognitive engagement is expected to promote large-scale functional integration and counteract age-related network segregation through coordinated recruitment of prefrontal–cerebellar and cortico–basal-ganglia loops (Doyon & Benali, 2005; Leisman et al., 2016). Aerobic-enriched sensorimotor activity upregulates hippocampal neurogenesis (Wollesen & Voelcker-Rehage, 2014), providing a substrate for the episodic memory gains that constitute the earliest marker of dementia risk. Bilateral midline-crossing movements — a structural feature of every Kinesthesia game — engage interhemispheric connectivity via the corpus callosum, and the proprioceptive–vestibular richness of full-body interaction may contribute independently to cerebellar plasticity and network synchronisation. At the synaptic level, the adequacy–reinforcement–frequency triad operationalised by the adaptive algorithm is expected to support long-term potentiation and functional network reorganisation (Nahum et al., 2013). All proposed mechanisms are theoretical constructs pending empirical validation; resting-state EEG was recorded in 19 of 20 participants and neurophysiological findings are reported separately as convergent evidence.

2.5 Outcome measures

Assessments were administered at baseline and within two weeks of the 30th session by trained evaluators blinded to individual game performance data. Primary cognitive outcomes were derived from the NEUROPSI Atención y Memoria battery (Ostrosky-Solís et al., 2007), which provides domain-specific measurement across subscales (orientation, attention, executive function, working memory, immediate and delayed memory) without ceiling effects, incorporates age- and education-stratified normative data, and has been validated in Spanish-speaking older adults. Primary motor outcome was the Six-Minute Walk Test (6MWT): total metres walked and theoretical percentage of the age-expected normative value (Enright equation). The Tinetti Performance-Oriented Mobility Assessment and Timed-Up-and-Go test were administered at baseline for sample characterisation only and are not included as repeated-measures outcomes.

Resting-state EEG was recorded pre- and post-intervention in 19 of 20 participants using a 19-channel cap (10–20 system, linked-earlobe reference). Spectral indices were extracted per standard quantitative EEG protocols and analysed as a secondary neurophysiological characterisation. Adherence (proportion of scheduled sessions completed) and post-intervention satisfaction (1–5 scale per game) were collected as primary feasibility outcomes.

2.6 Statistical analysis

Pre-to-post differences were assessed by non-parametric sign-flip permutation tests (5,000 permutations), appropriate for small samples with no distributional assumptions. Multiple comparisons were controlled by Benjamini–Hochberg (BH) false discovery rate correction applied separately within each instrument domain (NEUROPSI: 39 tests; 6MWT: 12 tests). Results with $q < 0.05$ were deemed statistically significant; $q = 0.05–0.10$ were classified as trends; uncorrected $p < 0.05$ but $q \geq 0.05$ are flagged as (*). Effect sizes are Cohen's d (mean delta / SD of deltas). An exploratory MoCA-gradient analysis used permutation-based Spearman rank correlations with BH correction within domain. Game–outcome correlations used BH correction within each game (see Supplementary Table S1 for full results). All findings are exploratory. NEUROPSI combined $N=20$; 6MWT combined $N=19$ (one MCI participant lacked a valid 6MWT).

3. Results

3.1 Adherence and feasibility

Adherence was 100% in both subgroups: all participants who reached the post-intervention assessment completed every scheduled session without dropout attributable to the intervention. No adverse events related to Kinesthesia were reported. Satisfaction ratings were uniformly high: MCI participants rated all games 4.25–5.0/5 (mean ≈ 4.75); HC participants rated all items above 4.5/5, with near-universal willingness to continue and recommend the programme. These results confirm high acceptability of Kinesthesia across both cognitive profiles.

3.2 Cognitive outcomes

The primary analysis (N=20, BH correction within 39 NEUROPSI tests) identified BH-significant improvements in eight variables ($q \leq 0.05$): spontaneous verbal recall ($d = +1.07$, $q = 0.004$), cued verbal recall ($d = +0.93$, $q = 0.004$), memory-learning curve ($d = +0.85$, $q = 0.014$), verbal recognition ($d = +0.65$, $q = 0.040$), associated pairs evocation ($d = +0.81$, $q = 0.023$), name evocation ($d = +0.75$, $q = 0.025$), sequential series ($d = +1.07$, $q = 0.010$), and global normalised score ($d = +0.93$, $q = 0.008$). Six additional variables showed BH trends ($q < 0.10$). Full data are presented in Table 1 and Figure 1.

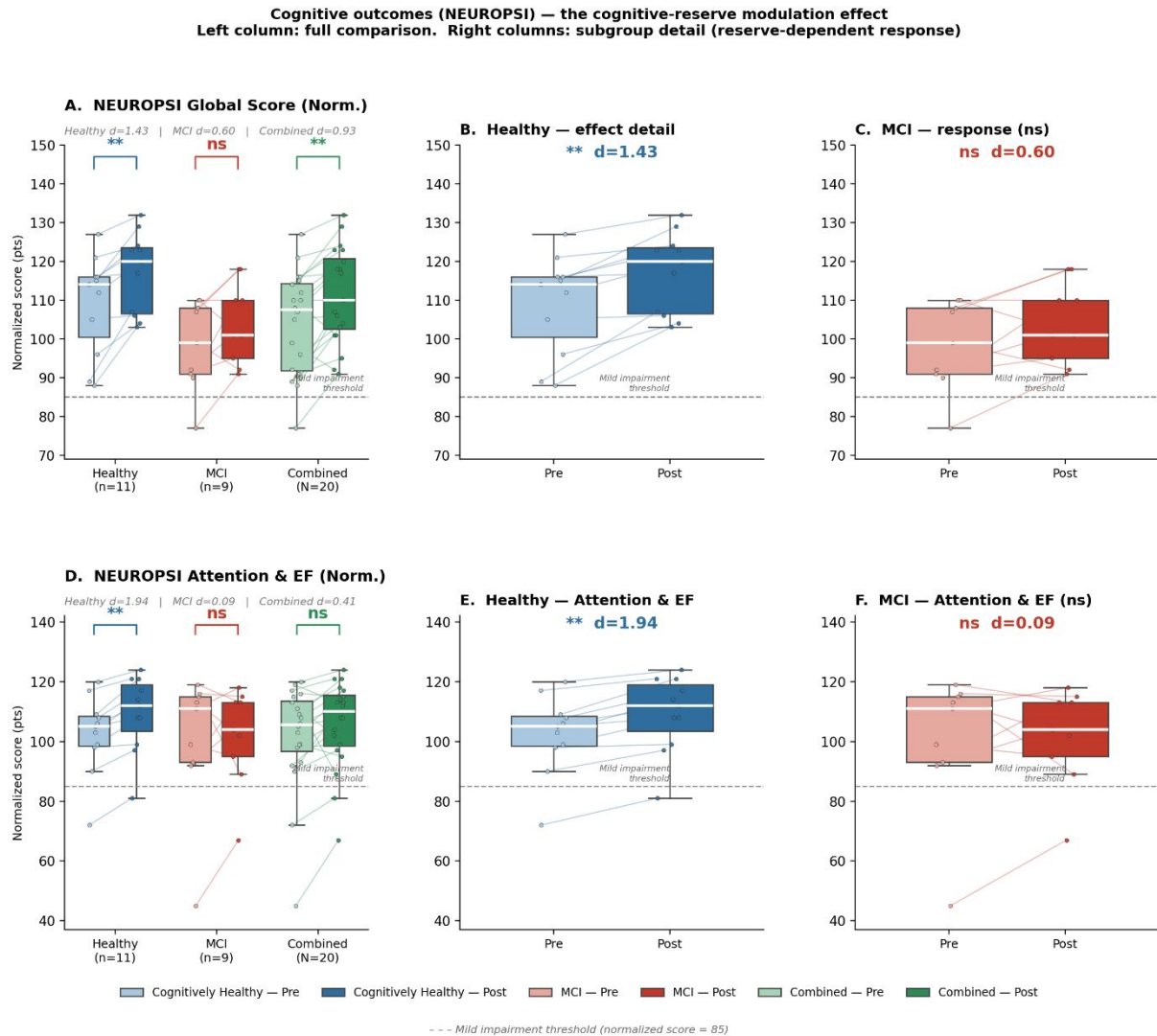


Figure 1. NEUROPSI cognitive outcomes — the cognitive-reserve modulation effect (NEUROPSI N=20; HC n=11, MCI n=9). Panels A and D show the full pre-to-post comparison for the global normalised score (A) and the Attention & Executive Function normalised composite (D) across the cognitively healthy (HC), MCI, and combined samples; panels B/C and E/F give the corresponding subgroup detail. Boxplots show pre (light) and post (dark) distributions with individual trajectories overlaid; the dashed line marks the mild-impairment threshold (normalised score = 85). Effect sizes are Cohen's d (Δ/SD of deltas). In the cognitively healthy group both composites improved with large, BH-significant effects (global $d = 1.43$, $q = 0.012$; Attention/EF $d = 1.94$), crossing from the “normal” into the “high-normal” range, whereas the MCI group showed no significant change (global $d = 0.60$, $p = 0.116$; Attention/EF $d = 0.09$)

— a reserve-dependent dissociation. At the combined level the global normalised score improved significantly ($d=0.93$, $p<0.001$, BH $q=0.008$), while the Attention/EF composite did not ($d=0.41$, $p=0.084$). All findings are exploratory; the study lacked a randomised control condition.

Alt Text: Six-panel figure showing pre-to-post changes in NEUROPSI cognitive scores. Top row: global normalised score for combined sample (A), cognitively healthy HC subgroup (B), and MCI subgroup (C). Bottom row: Attention and Executive Function composite (D, E, F). Each panel shows boxplots with individual trajectories overlaid; blue represents HC, red represents MCI. A dashed horizontal line marks the mild-impairment threshold at normalised score 85. HC shows large improvements crossing into the high-normal range; MCI shows no significant change.

Exploratory subgroup analyses revealed qualitatively distinct profiles. In the MCI subgroup ($n=9$), eight variables showed uncorrected significance ($p<0.05$), concentrating on verbal memory ($d=+1.29$ – 1.67) and executive sequencing ($d=+1.28$) — the domains most compromised in early cognitive decline. In the HC subgroup ($n=11$), two variables survived BH correction: normalised Attention/Executive score ($d=+1.94$, $q=0.012$) and global normalised score ($d=+1.38$, $q=0.012$), reflecting broad cognitive activation. This MCI–HC dissociation — deficit normalisation versus performance enhancement — is theoretically consistent with state-dependent mechanisms predicted by cognitive reserve theory (Khalaila et al., 2024; Liu et al., 2023). The exploratory MoCA-gradient analysis showed no variable survived BH correction for the MoCA–delta association, indicating that improvements were broadly distributed across the cognitive reserve continuum rather than selective for the most impaired participants.

3.3 Motor outcomes

The primary analysis identified BH-significant improvements in both 6MWT variables: metres walked ($d=+1.20$, $p<0.001$, $q=0.002$) and theoretical percentage of the age-expected norm ($d=+0.76$, $p=0.002$, $q=0.023$). Mean gain was $+60.2$ m (pre: 415.6 ± 75.1 m; post: 475.9 ± 58.5 m), substantially exceeding the minimally important difference of 14–30 m for this population. Subgroup analyses showed consistent direction: MCI $d=+1.40$ ($p=0.004$, $q=0.024$) and HC $d=+1.06$ ($p=0.006$, $q=0.024$), both BH-significant. The convergence of subgroup direction with the BH-confirmed combined result strengthens confidence in this finding (Figure 2).

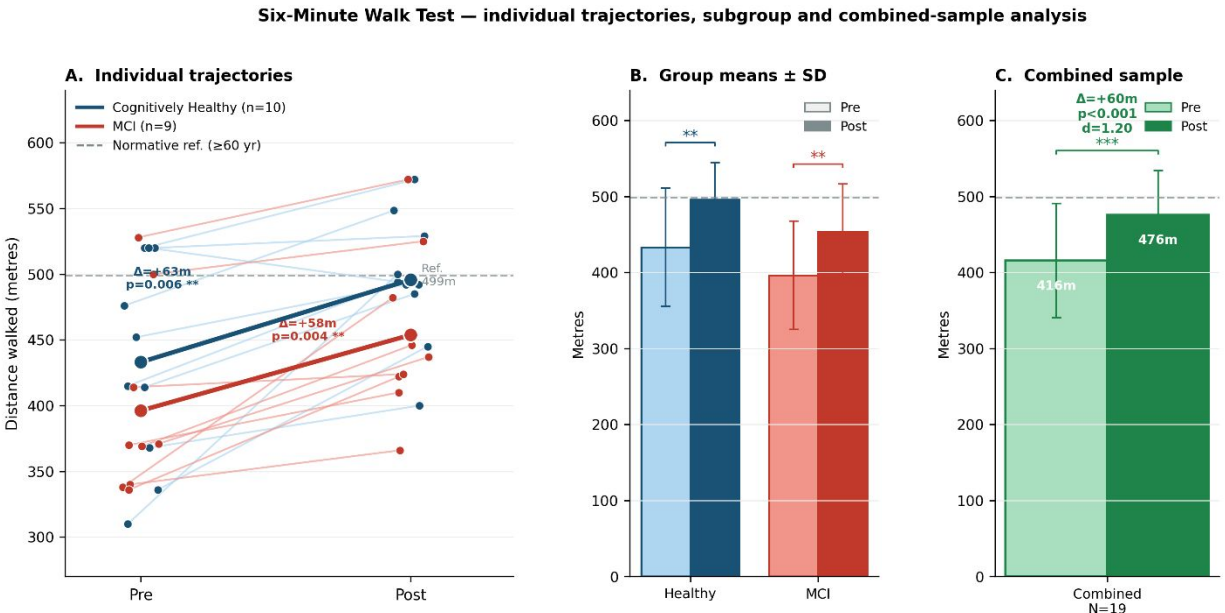


Figure 2. Six-Minute Walk Test — individual trajectories, subgroup, and combined-sample analysis (6MWT N=19; HC n=10, MCI n=9). (A) Individual pre-to-post trajectories (thin lines) with subgroup means (thick lines); blue = HC, red = MCI; the dashed line marks the age-expected normative reference (Enright equation). (B) Subgroup means $\pm$ SD, pre versus post. (C) Combined-sample means. Distance walked increased by +62.6 m in HC ($d=1.06$, $p=0.006$) and +57.6 m in MCI ($d=1.40$, $p=0.004$); the combined gain was +60.2 m ($d=1.20$, $p<0.001$, BH $q=0.002$), well above the 14–30 m minimally important difference for this population. All findings are exploratory; the study lacked a randomised control condition.

Alt Text: Three-panel figure showing Six-Minute Walk Test results. Panel A shows individual pre-to-post trajectories as thin lines with thick subgroup mean lines; blue for cognitively healthy HC, red for MCI; a dashed line marks the age-expected normative reference. Panel B shows subgroup means plus or minus SD pre versus post. Panel C shows combined-sample means. Both groups improved substantially, with combined gain of 60.2 metres above the minimally important difference threshold.

3.4 Neurophysiological correlates

Of 10 pre-specified MoCA-gradient spectral EEG indices, 8 survived BH correction ($q<0.05$) at baseline, confirming that pre-intervention neurophysiological state covaried with the cognitive reserve continuum. Spectral indices were evaluated against a multinational normative database spanning nine countries and approximately 2,000 participants, allowing individual deviations from age-expected norms to be quantified. Post-intervention, 21 of 23 pre-existing spectral differences between subgroups resolved, while 44 new spectral patterns emerged — a reorganisation characterised by reduction of slow-wave excess and emergence of alpha-range activity, consistent with spectral normalisation accompanying the observed behavioural improvements. Full EEG methodology and spectral index results are reported separately; neurophysiological findings are summarised here as convergent evidence only.

3.5 Within-platform learning and game–outcome associations

Session-by-session performance data demonstrated systematic skill acquisition across all analysed games: mean regression slopes for scores were positive and for execution times were negative, with 80–91% of participants showing individually significant improvement trajectories in the highest-exposure games (Categorization variants, Follow the Trail, Two Steps, Lit Tiles; $p < 0.001$). Spearman correlations between individual game learning indices and pre-to-post clinical change are reported in Supplementary Table S1. In the combined sample ($N = 14–18$ per game–outcome pair), 11 associations survived BH correction ($q < 0.05$), following three consistent patterns (Figure 3; full results in Supplementary Table S1). First, learning of Categorized N (attentional search game) predicted 6MWT improvement across the combined sample ($\rho = +0.79$, $q = 0.003$, $n = 17$) and Stroop inhibitory control gains in the MCI subgroup ($\rho = +0.93$, $q = 0.038$, $n = 7$), suggesting that attentional processing under locomotor load may constitute a shared substrate for both functional and inhibitory gains. Second, locomotor game learning (Follow the Trail, Lit Tiles) was inversely associated with 6MWT gains in HC (Follow the Trail: $\rho = -0.76$, $q = 0.015$; Lit Tiles: $\rho = -0.74$, $q = 0.020$; both $n = 11$), indicating that HC participants with the smallest locomotor learning curves — likely those already possessing efficient motor strategies — showed the greatest functional walking improvement, consistent with a transfer-from-mastery rather than a deficit-repair mechanism. Third, Follow the Trail learning in the MCI subgroup predicted category formation gains ($\rho = -0.93$, $q = 0.037$, $n = 7$), suggesting that foundational locomotor sequencing may scaffold higher-order categorical reasoning through shared frontoparietal recruitment in MCI. These domain-specific patterns are consistent with scaffolding theory (Park & Reuter-Lorenz, 2009), though correlational analyses cannot establish causality and small subgroup sizes ($n = 7–11$) limit interpretability.

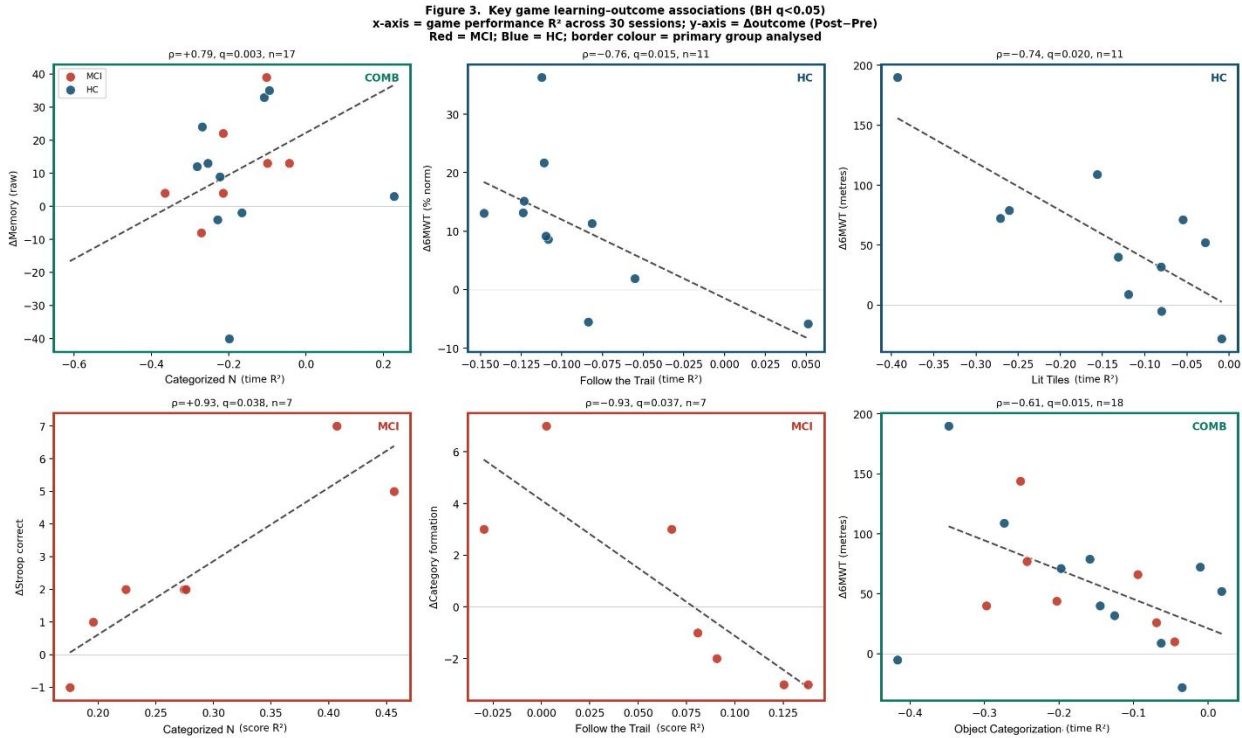


Figure 1: Key game learning-outcome associations surviving Benjamini-Hochberg FDR correction ($q < 0.05$). Each panel shows the Spearman correlation between individual game learning (x-axis: R^2 of performance trajectory across 30 sessions) and change in a clinical outcome (y-axis: Post-Pre). Red dots: MCI ($n=9$); blue dots: HC ($n=11$). Border colour indicates the primary group analysed: green = combined sample; red = MCI; blue = HC. Dashed line: linear fit for the analysed group. Spearman ρ and BH-corrected q -values are shown above each panel. Top row (combined sample and HC): higher learning in Categorized N (attentional search) was associated with greater memory gains in the combined sample ($\rho = +0.79$, $q = 0.003$, $n = 17$); lower learning rates in Follow the Trail and Lit Tiles (locomotor games) were associated with greater 6MWT gains in HC ($\rho = -0.76$, $q = 0.015$ and $\rho = -0.74$, $q = 0.020$, $n = 11$). Bottom row (MCI): higher Categorized N scores correlated with Stroop correct responses ($\rho = +0.93$, $q = 0.038$, $n = 7$); higher Follow the Trail scores correlated with category formation gains ($\rho = -0.93$, $q = 0.037$, $n = 7$). Full correlation results by game and subgroup are reported in Supplementary Table S1. All associations are exploratory; the study lacked a randomised control condition.

Alt Text: Six-panel scatter plot figure showing game learning-outcome correlations surviving Benjamini-Hochberg correction. Each panel plots individual game learning R-squared on the x-axis against pre-to-post clinical change on the y-axis. Red dots represent MCI participants; blue dots represent HC participants. Border colour indicates analysed group: green for combined, red for MCI, blue for HC. A dashed regression line is shown for the primary analysed group. Top row shows combined and HC associations; bottom row shows MCI associations.

Table 1. Pre-to-post cognitive and functional outcomes: effect sizes and Benjamini-Hochberg FDR correction.

Outcome	MCI (n=9)			HC (n=11)			Combined (N=20)					
	<i>d</i>	<i>p</i>	<i>q</i>	<i>d</i>	<i>p</i>	<i>q</i>	Pre M±SD	Post M±SD	Δ	<i>d</i>	p	q

NEUROPSI — Executive functions														
Sequential series	+1.28	0.029	0.164	(*)	+0.89	0.032	0.182	(*)	1.58±1.22	2.47±0.94	+0.89	+1.07	0.001	0.010 **
Stroop correct	+0.98	0.029	0.164	(*)	+0.28	0.424	0.688		—	—	—	+0.57	0.030	0.089 ~
Categorization	+0.23	0.609	0.911		+1.06	0.011	0.107	(*)	—	—	—	+0.63	0.015	0.053 ~
Norm. Attn/Exec.	+0.09	0.793	0.967		+1.94	<0.001	0.012	**	—	—	—	+0.41	0.084	0.204
Norm. global score	+0.05	0.867	—		+1.38	0.002	0.012	**	105.4±13.9	110.0±12.9	+4.6	+0.93	<0.001	0.008 **
NEUROPSI — Verbal memory														
Memory-learning curve	+1.67	0.007	0.137	(*)	+0.49	0.197	0.559		—	—	—	+0.85	0.002	0.014 **
Spont. verbal recall	+1.38	0.015	0.137	(*)	+0.85	0.033	0.182	(*)	6.63±2.75	8.26±2.22	+1.63	+1.07	<0.001	0.004 **
Cued verbal recall	+1.29	0.016	0.137	(*)	+0.69	0.073	0.315		—	—	—	+0.93	<0.001	0.004 **
Verbal recognition	+0.88	0.036	0.174	(*)	+0.53	0.171	0.559		—	—	—	+0.65	0.008	0.040 **
Assoc. pairs evoc.	+0.58	0.175	0.568		+1.07	0.015	0.120	(*)	8.16±2.9	9.74±2.4	+1.58	+0.81	0.004	0.023 **
Name evocation	+0.89	0.066	0.286		+0.69	0.071	0.315		—	—	—	+0.75	0.004	0.025 **
6-Minute Walk Test														
6MWT metres (m)	+1.40	0.004	0.024	*	+1.06	0.006	0.024	*	415.6±75.1	475.9±58.5	+60.2	+1.20	<0.001	0.002 **
6MWT % theoretical	+0.48	0.129	—		+1.11	0.006	0.024	*	77.6±13.9	87.8±12.4	+10.2	+0.76	0.002	0.023 *

d = Cohen's d (mean delta/SD of deltas); p = sign-flip permutation p -value (5,000 permutations); q = BH-corrected p -value. BH correction applied separately within NEUROPSI (39 tests) and 6MWT (12 tests). Flags: ** $q < 0.05$; ~ $q < 0.10$ trend; (*) $p < 0.05$ uncorrected only. Full pre-to-post means $\pm$ SD in Supplementary Table S2.

Table 2. Alignment of Kinesthesia with six neuroplasticity-based rehabilitation principles (Nahum et al., 2013) and proposed neurophysiological monitoring markers.

BP = Bereitschaftspotential; MCP = Motor Cortex Potential; SPP = Skilled Performance Positivity; PAN = Post-Action Negativity; MRP = Movement-Related Potential. For a review of MRP applications in cognitive rehabilitation monitoring, see (Chiarenza & Bosch-Bayard, 2025).

Principle	Definition	Implementation in Kinesthesia	Proposed MRP marker
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Attention	Training must engage sustained, focused attention; passive engagement is insufficient for plastic change	Simultaneous cognitive and motor demands require undivided attention throughout each task; attentional disengagement produces immediate performance failure, providing intrinsic motivation to sustain engagement	<i>BP amplitude; N100 (sensory attention gating)</i>
Accuracy	Responses must be spatially and temporally precise; imprecision must be detected and corrected	Full-body motion capture (centimetre resolution) enforces precise responses in every game; only accurate motor outputs are reinforced by the game narrative	<i>MCP (sensorimotor integration); P200 (motor completion)</i>
Adequacy	Difficulty must match individual capacity — neither too easy nor too demanding — and adapt continuously	Bidirectional adaptive algorithm adjusts stimulus speed, density, and spatial range within each game in real time; difficulty level is retained across sessions for each participant	<i>BP onset and amplitude (strategic motor programming)</i>
Reinforcement	Successful performance must be immediately and distinctively rewarded to gate synaptic strengthening	Real-time visual and auditory feedback on every correct response; emotionally engaging game narrative provides immediate reward; structured feedback distinguishes correct from incorrect responses	<i>SPP (performance outcome evaluation); PAN (error detection)</i>
Frequency	Sufficient repetition density and session frequency are required for consolidation	30 sessions × 60 minutes (≈30 min active play per session), 3×/week, 10 weeks; within-session repetition density and between-session spacing optimise consolidation	<i>BP amplitude reduction across sessions (motor engram formation)</i>
Flexibility	Programme must be individually adaptable and generalisable across settings	Modular 13-game suite; individualised session composition; applicable from normal aging to MCI; deployable in clinical, community, and low-resource settings	<i>SPP latency trajectory across sessions</i>

4. Discussion

This uncontrolled pilot provides preliminary feasibility and efficacy evidence for Kinesthesia in community-dwelling older adults spanning the healthy to MCI spectrum. The primary findings — BH-corrected improvements in eight verbal memory and global cognitive variables ($d=0.65\text{--}1.07$) and both 6MWT outcomes ($d=0.75\text{--}1.00$) across $N=20$, together with 100% adherence and high satisfaction — constitute the core result. The neurophysiological findings provide convergent evidence: post-intervention resolution of 21 of 23 pre-existing EEG spectral differences is consistent with neural reorganisation accompanying behavioural improvement. All findings are exploratory; causal attribution to the intervention requires a randomised controlled design.

Published meta-analytic benchmarks report effect sizes of $g=0.21\text{--}0.37$ on global cognition for combined cognitive–physical programmes in MCI (Karssemeijer et al., 2017; Wollesen & Voelcker-Rehage, 2014) and $d=0.3\text{--}0.6$ for exergame-based approaches (Gallou-Guyot et al., 2020). The present BH-confirmed effect sizes ($d=0.65\text{--}1.94$) exceed these benchmarks, though

uncontrolled pre-post designs systematically overestimate effects relative to randomised trials, and practice effects on neuropsychological instruments constitute a plausible contributor in the absence of a control condition. The 6MWT gain (+60.2 m across 10 weeks) substantially exceeds the minimally important difference of 14–30 m for this population, representing a clinically meaningful improvement in functional aerobic capacity and mobility — an outcome of direct relevance to fall prevention and independence maintenance in older adults.

The most theoretically informative secondary finding is the qualitatively distinct subgroup response. The MCI profile — largest gains in verbal memory ($d=+1.29$ – 1.67) and executive sequencing ($d=+1.28$) — maps onto the domains most compromised in early decline. The HC profile — largest gains in executive composite ($d=+1.94$) and global normalised performance ($d=+1.38$, crossing into the ‘high normal’ NEUROPSI range) — reflects broad cognitive activation in individuals without a specific deficit to repair. Cognitive reserve theory offers one explanatory framework (Khalaila et al., 2024; Liu et al., 2023): individuals with MCI may have chronically exceeded available reserve, and structured sensory-motor cognitive challenge may restore access to residual circuit capacity through Hebbian reinforcement of partially degraded networks — a repair mechanism. In HC, where reserve remains intact, the same challenge may recruit additional circuits beyond the habitual network — an expansion mechanism. These two processes predict distinct neural signatures: repair should produce spectral normalisation (convergence toward age-normative EEG patterns), while expansion should produce de novo spectral differentiation. The post-intervention EEG finding — resolution of 21 of 23 pre-existing between-subgroup spectral differences alongside emergence of 44 new patterns — is consistent with simultaneous repair in MCI and expansion in HC, though this interpretation requires direct testing with individual-level EEG trajectory analysis in a controlled design. The Scaffolding Theory of Aging and Cognition (Park & Reuter-Lorenz, 2009) provides complementary framing: the aging brain recruits scaffolding circuits proportionate to the gap between task demand and available capacity. Kinesthesia’s adaptive difficulty algorithm may be particularly effective because it creates a calibrated, state-dependent demand gap — larger in MCI (triggering repair-oriented scaffolding) and optimally challenging in HC (triggering performance enhancement). If replicated in a controlled design, this pattern would have direct implications for stratified clinical deployment: Kinesthesia may function simultaneously as a restorative intervention for MCI and a performance-optimising one for HC, with the same protocol producing different but complementary neuroplastic outcomes depending on baseline cognitive state.

The resting-state EEG findings provide convergent neurophysiological evidence: the majority of pre-specified spectral indices covaried with the cognitive reserve continuum at baseline, consistent with prior evidence that quantitative EEG tracks MCI-related neurophysiological change (Babiloni et al., 2009; Smailovic & Jelic, 2019). Evaluated against a multinational normative reference spanning nine countries, the post-intervention spectral reorganisation — characterised by reduction of slow-wave excess and emergence of alpha-range activity in regions previously showing pathological deviation — is consistent with spectral normalisation accompanying

behavioural improvement. This pattern aligns with the neuroplastic mechanisms hypothesised to underlie sensory-motor cognitive training (Chiarenza, 1982; Ali et al., 2022), though causal attribution requires a controlled design with an EEG comparator condition.

The domain-specific game–outcome correlations — locomotor games predicting walking capacity in HC; attentional and sequential ordering games predicting cognitive gains in MCI — suggest that individual game mechanics may engage and reinforce task-congruent neural circuits, consistent with scaffolding theory (Park & Reuter-Lorenz, 2009). However, with $n=7-11$ per subgroup these correlations are exploratory and may reflect shared baseline capacity rather than specific transfer.

4.1 Limitations

The principal limitation is the absence of a randomised control group: observed pre-to-post changes cannot be causally attributed to the Kinesthesia intervention. Practice effects on repeated neuropsychological administration, regression to the mean, therapist attention, and natural fluctuation in functional capacity may contribute substantially. The small sample ($N=20$; $n=9$ MCI, $n=11$ HC) limits statistical power: with $n=9-11$, the sign-flip test achieves approximately 55% power for $d=0.80$ at $\alpha=0.05$ with BH correction across 39 tests. The sample was recruited from a single community centre with marked gender imbalance (86.4% women); generalisability cannot be assumed. The substantial educational difference between subgroups (~ 6.3 years, $p<0.001$) is a potential confound for cognitive outcomes not fully resolved by MoCA stratification alone. MCI classification based on MoCA alone does not fully meet NIA-AA or DSM-5 criteria. Finally, the gap between data collection (2017) and publication reflects institutional and resource constraints rather than methodological concerns; the study protocol and analytical approach are consistent with current standards.

4.2 Implications for a confirmatory trial

The BH-confirmed effect sizes provide an empirical basis for RCT sample size estimation. For 80% power at $\alpha=0.05$, detecting $d=0.80$ (conservative verbal memory estimate) requires $n\geq 26$ per arm ($N\geq 52$ total); detecting $d=1.00$ (6MWT, combined) requires $n\geq 22$ per arm ($N\geq 44$). These estimates should be treated as upper bounds given the uncontrolled design. The confirmatory RCT should incorporate an active control condition (e.g., conventional dual-task walking programme or social exercise group), stratified randomisation by MoCA and educational level, pre-registered primary outcomes (delayed verbal memory composite; 6MWT metres walked), assessor blinding, and follow-up at 3 and 6 months to evaluate durability of gains. Beyond the confirmatory RCT, a longer-term research agenda encompasses: validation of remote supervision and home-based delivery modalities to confirm that clinical-grade outcomes are preserved outside supervised settings; extension to other populations sharing overlapping cortico-striato-cerebellar targets — including Parkinson's disease, post-stroke rehabilitation, and children with neurodevelopmental disorders — where the platform is already deployed in clinical practice; and prioritisation of low- and middle-income country settings, where the burden of cognitive decline is disproportionate,

pharmacological options remain inaccessible, and affordable scalable platforms are most urgently needed.

5. Conclusions

Kinesthesia demonstrated strong feasibility — 100% adherence and uniformly high satisfaction across both cognitive profiles — and large-effect improvement signals in verbal memory, global cognitive function, and functional walking capacity in a small pilot sample of community-dwelling older adults with and without MCI. Convergent neurophysiological findings support the behavioural signal. The MCI–HC dissociation — deficit normalisation versus performance enhancement — is the most theoretically informative secondary finding and constitutes the primary hypothesis for confirmatory evaluation. All findings are exploratory; proposed mechanisms remain theoretical pending replication in a randomised controlled trial with active comparators. The platform's scalability, low cost, and suitability for community and low-resource settings make it a relevant candidate for implementation research in contexts where the burden of cognitive decline is high and pharmacological options remain inaccessible.

CRedit authorship contribution statement

Jorge Bosch-Bayard: Conceptualization, Methodology, Formal analysis, Software, Visualization, Writing – original draft, Writing – review & editing, Supervision, Project administration. **Judith Guerrero-Sauzamedia:** Investigation, Data curation, Resources. **Laura Navarrete-Torres:** Investigation, Data curation, Resources. **Rodolfo Bosch-Bayard:** Conceptualization, Methodology, Investigation, Resources, Validation, Writing – review & editing. **Giuseppe A. Chiarenza:** Conceptualization, Validation, Writing – original draft, Writing – review & editing. **Liane Fabre-Aguilar:** Conceptualization, Methodology, Formal analysis, Supervision, Validation, Writing – review & editing. **René Valdés-Rodríguez:** Conceptualization, Methodology, Formal analysis, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Ethics approval

This study was approved by the Research Ethics Committee of the Universidad Autónoma de Querétaro ((protocols 02-03/11/2016 and 03-01/12/2016) and was conducted in accordance with the Declaration of Helsinki.

Informed consent

All participants provided written informed consent prior to enrolment.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used Claude (Anthropic) to assist with language editing, grammatical refinement, and improvement of the clarity and readability of the manuscript text in English. After using this tool, the authors reviewed and edited the content as needed and took full responsibility for the content of the published article.

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Kinesthesia — Supplementary Material *Bosch-Bayard et al., 2026*

Kinesthesia: feasibility and preliminary efficacy of a sensory-motor cognitive intervention in community-dwelling older adults with and without Mild Cognitive Impairment

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Supplementary Table S1. Full game–outcome Spearman correlation table

Spearman ρ between game learning index (R^2 of score or time regression across 30 sessions, whichever yields lower p) and pre-to-post clinical change, by subgroup. BH FDR correction applied within each game across all NEUROPSI + 6MWT outcomes (up to 14 tests per game per group).

Flags: ** $q < 0.05$ (BH-significant); ~ $q < 0.10$ (trend); — $n < 4$ (insufficient data). Blue shading: BH-significant. Green: trend. Row shading (blue): combined-sample result. All analyses exploratory.

Game	Grp	n	NEUROPSI outcomes (Spearman ρ)								6MWT		
			Mem curve	Verb spont	Verb cued	Verb recog	Assoc pairs	Name evoc	Stroop	Categ	Memory (raw)	6MWT (m)	6MWT (%)
Locomotor games													
Two Steps	MCI	7	-0.46	-0.80	-0.41	+0.38	+0.39	-0.56	+0.71	+0.68	-0.38	+0.43	+0.43
	HC	11	—	+0.27	-0.18	-0.68	-0.16	+0.21	+0.24	+0.11	-0.30	+0.31	-0.41
	All	18	+0.37	+0.37	+0.64~	-0.05	+0.37	-0.33	+0.11	-0.05	-0.28	+0.25	+0.15
Follow the Trail	MCI	7	+0.15	+0.71	+0.71	-0.47	-0.29	-0.09	-0.67	-0.93**	-0.86~	+0.11	-0.25
	HC	11	+0.72	+0.72	+0.20	+0.40	-0.40	+0.57	+0.33	+0.19	-0.14	-0.47	+0.72
	All	18	-0.17	-0.17	+0.35	-0.20	-0.35	+0.20	+0.18	-0.37	-0.41	-0.10	-0.26
Lit Tiles	MCI	7	+0.66	+0.66	-0.60	+0.14	+0.25	-0.72	+0.30	+0.58	+0.21	+0.75	-0.32
	HC	11	-0.47	-0.47	-0.41	+0.29	-0.26	+0.68	+0.46	-0.33	+0.28	+0.13	-0.74**
	All	18	-0.15	-0.15	-0.40	-0.06	+0.16	+0.10	+0.23	+0.21	+0.22	-0.17	+0.05
Attentional search													
Categorization	MCI	7	-0.58	-0.58	-0.40	+0.29	-0.55	+0.75	+0.11	+0.29	+0.18	-0.73	-0.61
	HC	11	+0.22	+0.22	+0.58	+0.33	+0.25	-0.16	-0.60	-0.32	+0.25	+0.41	+0.44
	All	18	-0.29	-0.29	-0.17	-0.05	-0.40	-0.26	+0.29	+0.43	+0.22	-0.37	-0.61**
Categorized N	MCI	7	+0.31	+0.31	+0.82	-0.79	-0.33	-0.04	+0.93**	-0.51	+0.61	+0.51	-0.68
	HC	11	-0.26	-0.26	-0.03	+0.23	-0.14	+0.61	-0.58	+0.24	-0.38	-0.04	+0.14
	All	17	+0.40	+0.40	+0.20	-0.13	-0.08	-0.11	+0.36	-0.21	+0.07	+0.79**	-0.29
Categorized 6	MCI	7	-0.46	-0.46	+0.56	-0.20	-0.33	+0.53	-0.33	-0.78	-0.46	-0.78	+0.71
	HC	11	-0.29	-0.29	+0.32	-0.39	+0.18	-0.22	-0.28	-0.23	+0.18	-0.32	-0.23
	All	18	+0.36	+0.36	+0.59	+0.21	+0.39	+0.39	-0.09	-0.37	-0.21	+0.29	+0.20
Sequential ordering													
Ascending Sort	MCI	7	-0.93**	-0.93**	-0.20	+0.58	+0.35	-0.75	+0.41	-0.13	+0.07	+0.84	+0.36
	HC	11	+0.19	+0.19	+0.25	-0.23	+0.29	+0.48	-0.06	-0.24	-0.12	+0.21	-0.52
	All	18	-0.28	-0.28	+0.16	-0.22	+0.08	+0.14	-0.53	+0.04	+0.04	+0.15	-0.23
Arithmetic													
Math Level 1	HC	8	-0.29	-0.29	+0.53	+0.60	-0.44	+0.27	-0.06	-0.91**	+0.17	-0.49	-0.43
Math Level 3	HC	7	+0.15	+0.15	+0.33	+0.03	+0.19	-0.06	+0.18	-0.94**	+0.50	-0.49	+0.66

Supplementary Table S2. Pre-to-post means $\pm$ SD across all outcome domains

Pre-to-post means $\pm$ SD, Wilcoxon signed-rank p-values, and Cohen's d for all analysed outcome variables by subgroup and combined sample (N=20: n=11 HC, n=9 MCI).

Flags: *** p<0.001; ** p<0.01; * p<0.05; ~ p<0.10; ns = not significant. Effect size d = mean delta / SD of deltas (paired). Subgroup p-values are Wilcoxon signed-rank; combined-sample p-values and effect sizes are the sign-flip permutation values from the primary analysis and are identical to those reported in Table 1 of the main manuscript. Blue shading: combined-sample BH-significant or trend.

Outcome measure	Group	Pre M $\pm$ SD	Post M $\pm$ SD	Δ	p	Sig	d	Effect
6-Minute Walk Test								
6MWT Metres (m)	HC (n=11)	433.1 $\pm$ 77.7	495.8 $\pm$ 49.0	+62.6	0.006	**	1.06	LARGE
	MCI (n=9)	396.2 $\pm$ 71.3	453.8 $\pm$ 63.0	+57.6	0.004	**	1.40	LARGE
	Combined (N=20)	415.6$\pm$75.1	475.9$\pm$58.5	+60.2	<0.001	***	1.20	LARGE
6MWT % Theoretical	HC (n=11)	78.4 $\pm$ 15.0	90.9 $\pm$ 8.6	+12.5	0.006	**	1.11	LARGE
	MCI (n=9)	76.7 $\pm$ 13.5	84.3 $\pm$ 15.4	+7.6	0.129	ns	0.48	MED
	Combined (N=20)	77.6$\pm$13.9	87.8$\pm$12.4	+10.2	0.002	**	0.76	LARGE
NEUROPSI — Global scores								
Global Norm. Score	HC (n=11)	108.3 $\pm$ 13.4	116.5 $\pm$ 10.8	+8.2	0.002	**	1.38	LARGE
	MCI (n=9)	102.2 $\pm$ 14.5	102.8 $\pm$ 11.5	+0.6	0.867	ns	0.05	NEGL
	Combined (N=20)	105.4$\pm$13.9	110.0$\pm$12.9	+4.6	<0.001	***	0.93	LARGE
Attn/EF Norm. Score	HC (n=11)	102.1 $\pm$ 13.8	109.0 $\pm$ 13.5	+6.9	0.002	**	1.94	LARGE
	MCI (n=9)	104.2 $\pm$ 14.4	103.7 $\pm$ 10.0	-0.6	1.000	ns	0.04	NEGL
	Combined (N=20)	103.1$\pm$13.7	106.5$\pm$12.0	+3.4	0.084	ns	0.41	SMALL
NEUROPSI — Verbal memory								
Spont. Verbal Recall	HC (n=11)	8.50 $\pm$ 2.17	9.80 $\pm$ 1.48	+1.30	0.039	*	0.85	LARGE
	MCI (n=9)	4.56 $\pm$ 1.88	6.56 $\pm$ 1.59	+2.00	0.063	~	1.38	LARGE
	Combined (N=20)	6.63$\pm$2.75	8.26$\pm$2.22	+1.63	<0.001	***	1.07	LARGE
Assoc. Pairs Evoc.	HC (n=11)	10.1 $\pm$ 1.9	10.9 $\pm$ 1.1	+0.80	0.094	~	1.07	LARGE
	MCI (n=9)	6.00 $\pm$ 2.3	8.44 $\pm$ 2.9	+2.44	0.070	~	0.58	MED
	Combined (N=20)	8.16$\pm$2.9	9.74$\pm$2.4	+1.58	0.004	**	0.81	LARGE
Rey Figure Evoc.	HC (n=11)	13.5 $\pm$ 5.9	16.8 $\pm$ 7.4	+3.30	0.016	*	1.03	LARGE
	MCI (n=9)	10.0 $\pm$ 6.4	10.6 $\pm$ 6.0	+0.56	0.828	ns	0.19	NEGL
	Combined (N=20)	11.8$\pm$6.2	13.8$\pm$7.3	+2.00	0.013	*	0.64	MED
Cued Verbal Recall	HC (n=11)	8.00 $\pm$ 1.83	9.00 $\pm$ 1.63	+1.00	0.062	~	0.69	MED
	MCI (n=9)	5.44 $\pm$ 1.33	6.33 $\pm$ 1.73	+0.89	0.406	ns	1.29	LARGE
	Combined (N=20)	6.79$\pm$1.99	7.74$\pm$2.09	+0.95	<0.001	***	0.93	LARGE
NEUROPSI — Attention & Executive Function								
Successive Series	HC (n=11)	1.60 $\pm$ 1.35	2.60 $\pm$ 0.84	+1.00	0.062	~	0.89	LARGE

	MCI (n=9)	1.56±1.13	2.33±1.00	+0.78	0.359	ns	1.28	LARGE
	Combined (N=20)	1.58±1.22	2.47±0.94	+0.89	0.001	**	1.07	LARGE
Category Formation	HC (n=11)	14.2±4.4	17.1±4.6	+2.90	0.023	*	1.06	LARGE
	MCI (n=9)	10.0±2.8	10.8±4.1	+0.78	0.582	ns	0.23	SMALL
	Combined (N=20)	12.2±4.2	14.1±5.4	+1.90	0.015	*	0.63	MED

Supplementary Materials S3. The Kinesthesia Platform

S3. Kinesthesia Game Suite

Table S3a presents the complete Kinesthesia game suite with targeted cognitive domains, specific motor demands, and primary dual-task configuration for each game. Motor demands common to all games — spatial displacement, dynamic postural balance, bilateral limb coordination, and body-midline crossing — are present throughout the entire suite. Games are ordered from lowest to highest cognitive-motor complexity as applied in the Kinesthesia protocol progression. The adaptive difficulty algorithm adjusts parameters (stimulus speed, spatial range, response windows, stimulus density) in real time within each game; the ordering shown reflects the cross-session progression from foundational to complex integration games.

Abbreviations: EF = Executive Functions; ATT = Attention & Concentration; MEM = Memory; VSP = Visuospatial Processing; CALC = Numerical Calculation; CAT = Categorization; LAT = Laterality; WM = Working Memory.

Game	Cognitive Domains	Specific Motor Demands	Primary Dual-Task Type
Square Hunt	ATT (selective, sustained), EF (response inhibition), VSP	Rapid whole-arm reaching to precise spatial targets; stepping; rapid direction changes	Simultaneous visuomotor tracking under attentional load
Star Rain	ATT (divided, sustained), EF (action planning), VSP	Continuous bilateral upper-limb interception; lateral stepping; arm-foot coordination	Divided attention with continuous movement
The Skier	ATT (sustained), EF (anticipation, trajectory planning), VSP	Whole-body lateral displacement; anticipatory postural adjustments; dynamic balance	Spatial navigation under sustained attentional load
Mathematical Calculation	ATT, CALC, MEM (WM), VSP	Targeted arm reaching; foot placement to correct positions; postural control	High cognitive load simultaneous with precise motor response
3D Geometry	ATT, EF (abstract reasoning, mental rotation), MEM, VSP	Precise arm positioning; body orientation changes; midline crossing	Visuospatial abstraction simultaneous with motor precision
Maze Navigation	ATT, EF (spatial planning, cognitive flexibility), VSP	Continuous whole-arm reaching along maze walls; midline crossing; trunk rotation	Strongest interhemispheric demand in suite
Categorization Games	ATT, EF (cognitive flexibility, rule-based classification), CAT, VSP	Whole-body movement to designated spatial categories; stepping; response inhibition	Simultaneous semantic decision-making and whole-body motor response
Numerical Sequencing (hands)	MEM (sequential WM), EF (planning, sequencing), ATT, VSP	Sequential bilateral hand reaching in memorized order; midline crossing	Sequential WM load simultaneous with motor sequencing
Lit Tiles	ATT (sustained, sequential), EF (sequencing), LAT, VSP	Foot placement on illuminated tiles in sequence; laterality stepping; trunk rotation	Simultaneous laterality judgment and precise lower-limb execution
Two Steps	ATT (sustained, divided), EF (strategic planning), LAT, VSP	Independent bilateral lower-limb dissociation to separate targets; asymmetric weight transfer	Maximum motor-cognitive integration; dissociated bilateral lower-limb control

Inside the Rubik Cube	ATT, EF (motor planning, timing), LAT, VSP	Simultaneous upper- and lower-limb coordination; four-limb integration; high balance	Maximum four-limb independent coordination under executive planning load
Follow the Path	MEM (immediate spatial memory), ATT, EF (planning), VSP	Full-body spatial navigation reproducing a memorized path; sequential stepping	Spatial memory recall driving whole-body motor execution
Numerical Sequencing (displacement)	MEM (WM under locomotor load), EF, ATT, VSP	Full-body locomotion to numerical targets in correct sequence	Classic locomotor-cognitive dual-task: gait under WM load

S3c. Comparison with existing cognitive–motor intervention systems

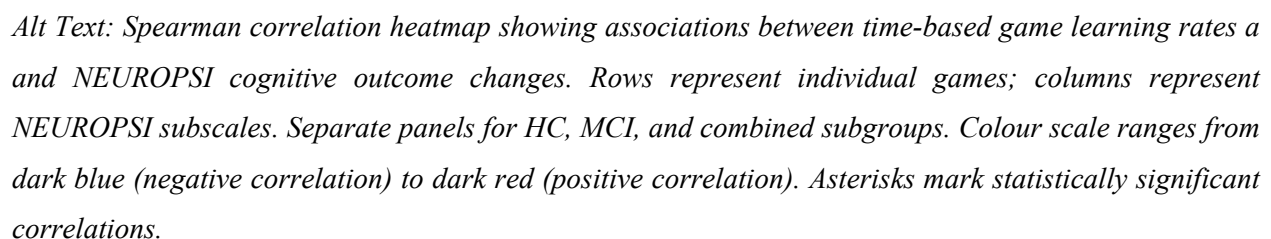
Table S3c compares Kinesthesia with three main categories of existing cognitive–motor intervention systems on nine design features. ✓ = feature present; ○ = partially present; X = feature absent. Sequential DT = sequential dual-task programmes; Adapted exergames = commercial games adapted for rehabilitation; Research VR = custom virtual reality systems.

Design Feature	Sequential DT	Adapted exergames	Research VR	Kinesthesia
Cognitive & motor demands simultaneous by design (not alternated)	X	○	○	✓
Systematic body-midline crossing	X	X	○	✓
All 6 neuroplasticity principles operationalised concurrently [21]	X	X	X	✓
Bidirectional adaptive difficulty algorithm	○	○	✓	✓
Clinically designed (not adapted entertainment software)	X	X	○	✓
Low hardware cost / standard computing equipment	✓	✓	X	✓
Scalable to community and home settings	✓	✓	X	✓
Session-level performance database per participant	X	X	○	✓
Validated across cognitive spectrum (healthy to MCI)	○	○	○	✓

Supplementary Figures S4–S7. Spearman correlation heatmaps: game learning vs clinical outcomes

Figures S4–S7 present full subgroup-level Spearman correlation heatmaps between within-platform game learning (R^2 of linear regression across 30 sessions) and pre-to-post clinical change (Δ POST–PRE). Two metrics are shown: score-based learning (R^2_{score} , reflecting improvement in task accuracy) and time-based learning (R^2_{time} , reflecting improvement in execution speed). Analyses are presented separately for the cognitively healthy subgroup (HC, $n=11$ game-matched), the MCI subgroup ($n=9$ game-matched), and the combined sample ($N=14–18$ per game). Only games with at least five participants showing a statistically significant learning slope ($p<0.05$) are included as rows. Colour encodes Spearman ρ : blue = positive; red = negative. Asterisks: * $p<0.05$, ** $p<0.01$. Non-significant cells are shown in white.

Supplementary Figure S₂. Spearman ρ : Game R² (Time — speed learning) vs Δ NEUROPSYCH.
Subgroup analyses (CS, MCI) and combined. $n \geq 5$, $p < 0.05$. Non-significant cells = white.



Supplementary Figure S₂. Spearman ρ : Game R² (Points — score learning) vs Δ NEUROPSI
Subgroup analyses (CS, MCI) and combined. $n \geq 5$, $p < 0.05$. Non-significant cells = white.

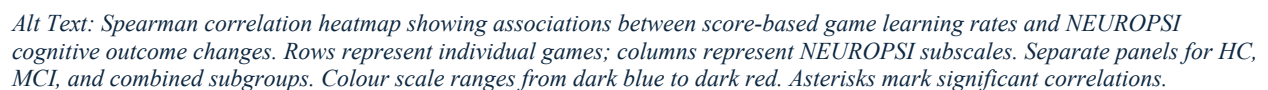
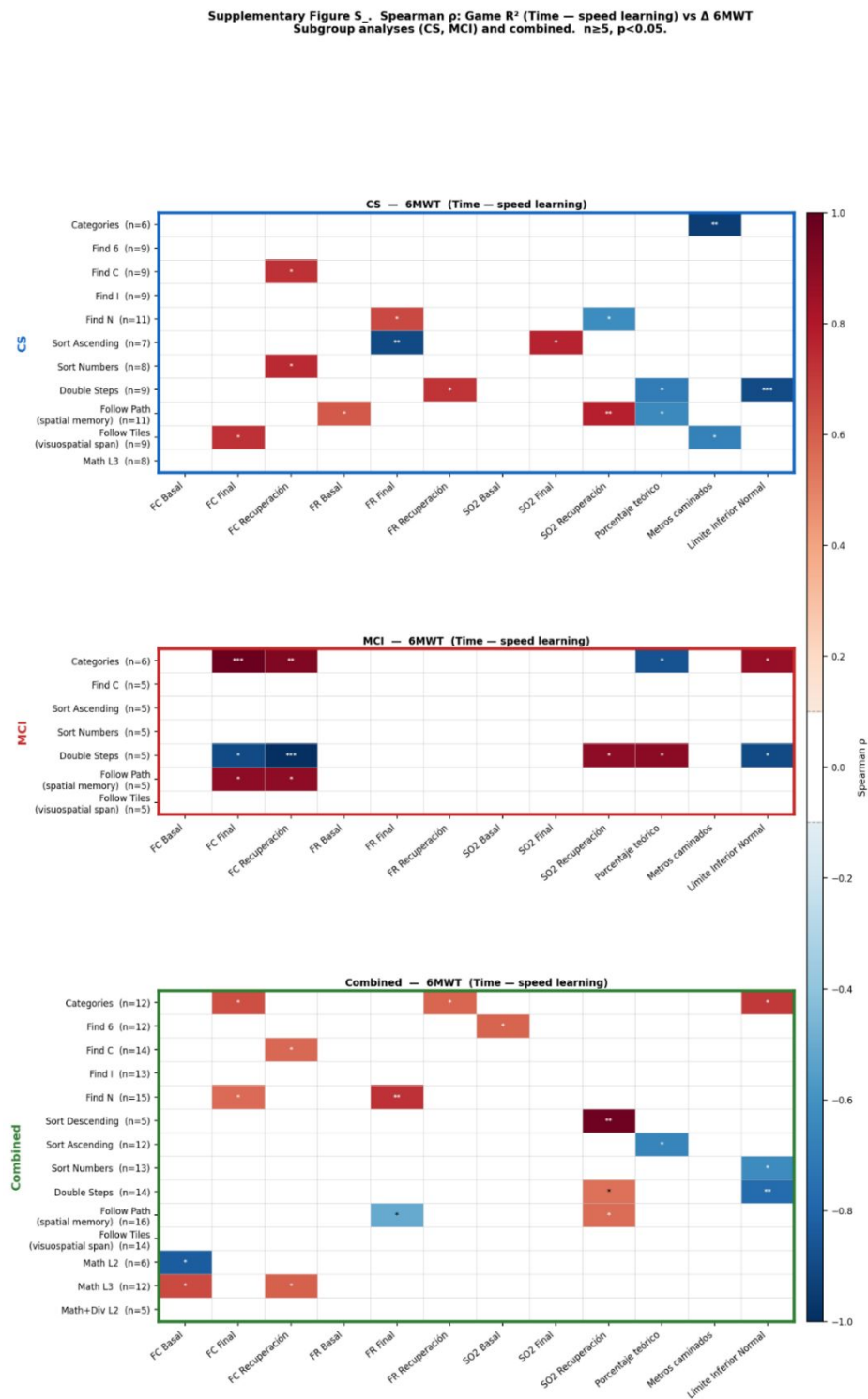
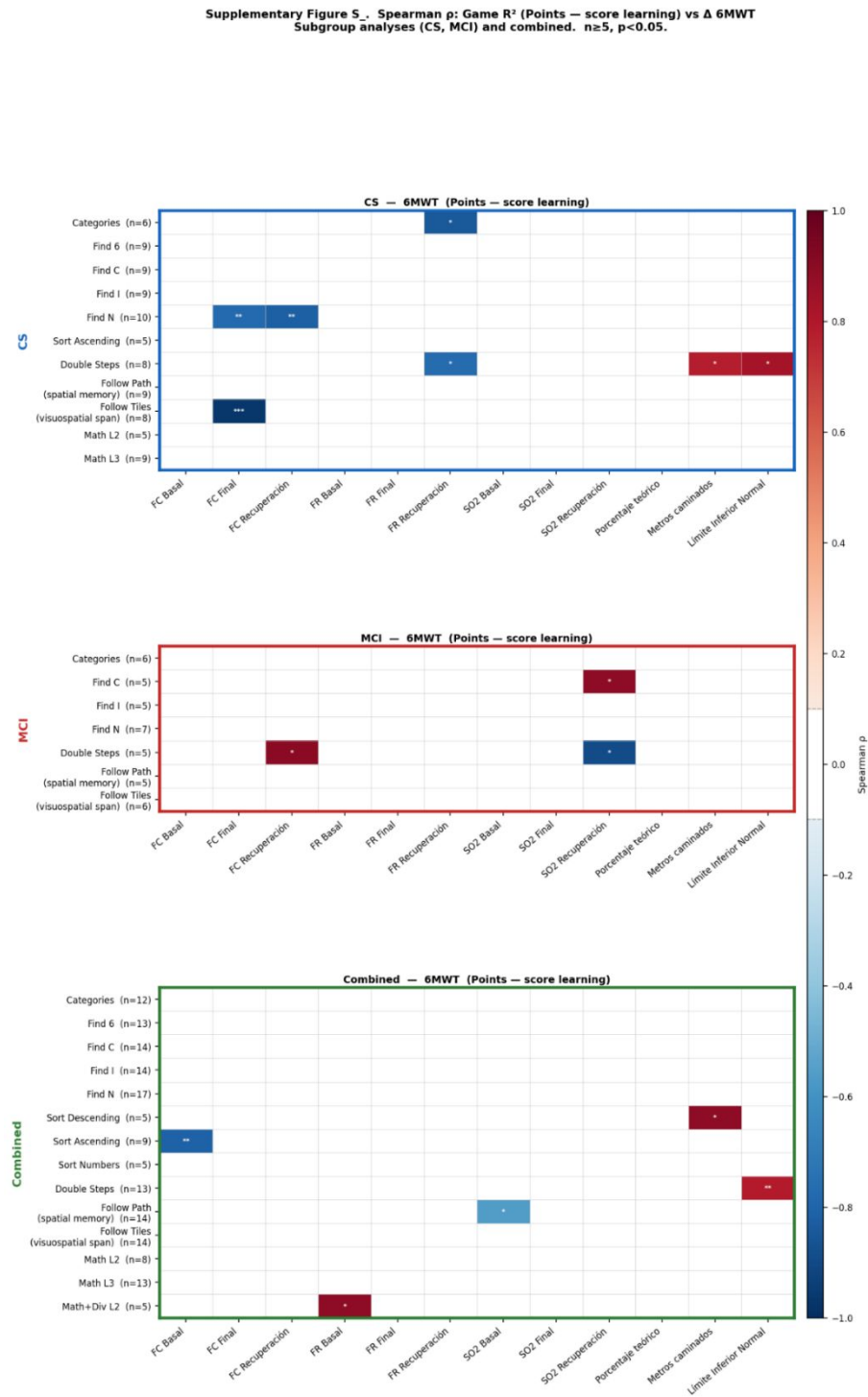


Figure S6. 6MWT — Time-based learning (R^2_{time}) $\times$ Δ 6MWT. HC, MCI, and combined subgroups.



Alt Text: Spearman correlation heatmap showing associations between time-based game learning rates and Six-Minute Walk Test outcome changes. Rows represent individual games; columns represent 6MWT variables. Separate panels for HC, MCI, and combined subgroups. Colour scale from dark blue to dark red; asterisks indicate significant correlations.

Figure S7. 6MWT — Score-based learning (R^2_{score}) $\times$ Δ 6MWT. HC, MCI, and combined subgroups.



Alt Text: Spearman correlation heatmap showing associations between score-based game learning rates and Six-Minute Walk Test outcome changes. Rows represent individual games; columns represent 6MWT variables. Separate panels for HC, MCI, and combined subgroups. Colour scale from dark blue to dark red; asterisks indicate significant correlations.

Kinesthesia: feasibility and preliminary efficacy of a sensory-motor cognitive intervention in community-dwelling older adults with and without Mild Cognitive Impairment

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Abstract

Mild Cognitive Impairment (MCI) represents a critical window for non-pharmacological intervention, yet accessible platforms combining cognitive and sensory-motor training remain scarce. We evaluated the feasibility and preliminary efficacy of Kinesthesia - a purpose-built, non-immersive therapeutic exergaming platform comprising serious exergames controlled through full-body movement - in 20 community-dwelling older adults spanning the cognitively healthy (n=11) to MCI (n=9) spectrum (mean age 69.7±3.6 years; 85% women). In this single-arm pre-post pilot, participants completed 30 sessions (60 minutes, three times weekly, over ten weeks). Cognitive and functional outcomes were assessed with the NEUROPSI Attention and Memory battery and the Six-Minute Walk Test; pre-to-post change was evaluated by sign-flip permutation testing with Benjamini-Hochberg correction, with an exploratory MoCA-gradient analysis. Adherence was 100%. BH-corrected improvements emerged across verbal-memory and executive indices (Cohen's d up to approximately 1.1) and in both walking outcomes (combined d=1.20). In exploratory subgroup analyses, the normalised global score improved in the cognitively healthy subgroup (d=1.38), whereas uncorrected MCI gains were concentrated in verbal memory;

between-group differences in change were not formally tested. These preliminary, hypothesis-generating findings support a confirmatory randomised controlled trial.

Keywords: mild cognitive impairment; cognitive rehabilitation; cognitive-motor training; exergaming; neuroplasticity; feasibility study

1. Introduction

The gap between life expectancy and disability-free years — approximately 10 years in high-income countries and proportionally larger in low- and middle-income settings — is driven primarily by cognitive decline, including Mild Cognitive Impairment (MCI) and dementia (Livingston et al., 2020; World Health Organization, 2015). MCI represents a critical preclinical window: approximately 50–70% of affected individuals progress to dementia within 5–7 years, though reversion to normal cognition is also documented, making early intervention both feasible and potentially high-impact (Menéndez González, 2014; Mitchell & Shiri-Feshki, 2009). In the absence of curative pharmacological treatments, international consensus supports multicomponent, non-pharmacological strategies combining physical activity, cognitive stimulation, and social engagement (Alzheimer’s Association, 2023; Livingston et al., 2020).

A central insight from contemporary neuroscience is that cognitive and motor functions are deeply interdependent, sharing overlapping neural substrates across frontoparietal, basal ganglia, thalamic, and cerebellar systems (Doyon & Benali, 2005; Leisman et al., 2016). This interdependence provides the mechanistic rationale for simultaneous sensory-motor cognitive interventions: co-activation of shared networks may induce broader neuroplastic changes than either cognitive or physical training alone. Meta-analyses of combined cognitive-physical programmes confirm improvements in executive function, balance, and gait in MCI, with simultaneous protocols generally outperforming sequential approaches (Karssemeijer et al., 2017; Wollesen & Voelcker-Rehage, 2014). Exergame-based interventions similarly show positive effects on cognition and functional mobility (Gallou-Guyot et al., 2020). An exergame is a digital game in which voluntary bodily movement is required to control gameplay and is integral to task performance. Two independent dimensions are important when interpreting this heterogeneous literature. By development purpose, systems may use commercial off-the-shelf recreational games subsequently repurposed for rehabilitation or purpose-built serious exergames designed around predefined health or clinical aims. By immersion level, either type may be non-immersive, using a conventional screen and motion tracking, or fully immersive, using a head-mounted display (Manser et al., 2025; Cieřlik, 2025). These configurations differ in content specificity, hardware, supervision, tolerability, and implementation requirements. Recent comparative syntheses do not show a monotonic relationship between greater immersion and greater cognitive efficacy (Hu et al., 2026; Qin et al., 2025).

Neurophysiological evidence further supports the mechanistic plausibility of such approaches: EEG studies show that sensory-motor cognitive dual-task protocols produce measurable changes in movement-related cortical potentials and alpha/beta event-related desynchronisation alongside improvements in inhibitory control (Chiarenza, 1982; Chiarenza & Bosch-Bayard, 2025; Ali et al., 2022). Quantitative EEG indices have demonstrated clinical utility as biomarkers along the healthy-to-MCI-to-dementia continuum (Babiloni et al., 2009; Smailovic & Jelic, 2019), offering a neurophysiological lens through which intervention-induced plasticity can be characterised.

In this context, we developed Kinesthesia - a purpose-built, non-immersive therapeutic exergaming platform comprising 13 serious exergames across five cognitive-motor categories¹. Kinesthesia uses a conventional display and commercially available full-body motion-capture hardware rather than a head-mounted VR device. Portability, accessibility, and deployability across clinical, community, and home-based settings, including low-resource contexts, are deliberate engineering characteristics of the platform. The present pilot did not evaluate total implementation cost, cost-effectiveness, staff burden, or real-world scaling outcomes. This paper reports feasibility and preliminary pre-to-post efficacy data in 20 community-dwelling older adults spanning the healthy to MCI spectrum, including brief resting-state EEG findings recorded in the same sample. A brief proof-of-concept communication summarising primary outcomes from this dataset has been published as a Letter to the Editor (Bosch-Bayard et al., 2026a); the present manuscript constitutes the complete scientific report with comprehensive methodology, full statistical characterisation, and extended theoretical analysis.

2. Methods

2.1 Study design

We conducted a single-arm, pre-post pilot study with repeated measures. The primary analysis treated participants as a single continuous sample (N=20), using the Montreal Cognitive Assessment (MoCA) score as a continuous index of cognitive reserve. Exploratory subgroup comparisons (HC: MoCA $\geq$ 26; MCI: MoCA<26) are reported as secondary evidence. All procedures complied with the Declaration of Helsinki; ethics approval was granted by the Ethics Committee of the Faculty of Medicine, Universidad Autónoma de Querétaro (protocols 02-03/11/2016 and 03-01/12/2016; institutional grant FOPER No. 143834).

¹ Platform-status note. The 13-game description refers to the version evaluated in 2017. Kinesthesia now includes more than 30 games at graded levels of complexity; approximately half can run online with a standard webcam, providing the technical basis for software-guided home sessions. Three-dimensional games retain Kinect 360 depth sensing. In September 2026, a developer price check, replacement Kinect 360 units were available on Amazon, eBay, Mercado Libre, and other online marketplaces for under US\$50, whereas technically suitable Intel RealSense alternatives were listed from approximately US\$400, with many configurations priced higher. This cost difference, the team's failure-free experience with its Kinect units over more than ten years, adequate three-dimensional tracking, and the mature SDK explain the deliberate decision to retain the sensor. Hardware communication is isolated in one module, permitting later sensor migration after adaptation and validation without redesigning the games. The price comparison is a dated descriptive snapshot, not a formal market or cost-effectiveness analysis; supervised home delivery and real-world scaling remain to be evaluated.

2.2 Participants

Seventy older adults attending the Plan Vida community day centre in Querétaro, Mexico, were screened in February 2017. Inclusion criteria: age ≥ 60 years; independent ambulation without assistive devices; no uncontrolled diabetes mellitus or arterial hypertension; no active cardiopulmonary disease; absence of neurodegenerative disease, thyroid disorder, or anaemia; absence of clinically significant depression (Yesavage Geriatric Depression Scale <5); no hazardous alcohol use (AUDIT <8); no moderate-to-severe uncorrected sensory impairment; and attainment of a minimum 6MWT threshold for safety. Of 70 screened, 22 met all criteria and enrolled; of these, 20 had complete pre- and post-intervention outcome data ($n=11$ HC, $n=9$ MCI) and constitute the analysis sample. Two MCI participants did not complete post-intervention assessment for non-intervention-related reasons.

The sample comprised 86.4% women (19 of 22 enrolled; the proportion among the 20 analysed was 85%), mean age 69.7 ± 3.6 years (range 62–76), MoCA range 15–29 (mean 24.8 ± 3.7). Only one participant scored below 17 (MoCA=15). HC participants had a mean age of 70.1 ± 3.8 years and 13.9 ± 2.5 years of education; MCI participants had a mean age of 69.3 ± 3.4 years and 7.6 ± 3.0 years of education. The subgroups differed substantially in education ($t=4.92$, $p<0.001$). Spearman correlation between MoCA and education was $\rho=0.50$ ($p=0.024$), driven predominantly by between-group differences. MoCA was retained as the sole cognitive reserve covariate to avoid multicollinearity ($N=20$); educational stratification is planned for the confirmatory RCT.

Cognitive status was established using the MMSE for initial eligibility screening and the MoCA (Spanish version; MCI threshold <26) for subgroup classification (Nasreddine et al., 2005). Neither instrument was used as a repeated-measures outcome variable, given well-documented ceiling effects and limited sensitivity to short-term change (Franco-Marina et al., 2010; Malek-Ahmadi et al., 2019). MoCA is a screening rather than a diagnostic instrument; therefore, the HC and MCI labels in this study denote operational subgroup classifications rather than formal clinical diagnoses. The participant with MoCA=15 may have had a level of impairment exceeding that ordinarily inferred from a formally diagnosed MCI sample. Classification based solely on MoCA, without functional assessment for diagnostic adjudication, full neuropsychological confirmation, or neuroimaging, does not meet current NIA-AA or DSM-5 diagnostic criteria (Albert et al., 2011; American Psychiatric Association, 2013) and limits interpretation of subgroup findings.

2.3 The Kinesthesia platform

The version of Kinesthesia evaluated in this 2017 study was a purpose-built, non-immersive therapeutic exergaming platform comprising 13 serious exergames across five cognitive-motor categories. An exergame is a digital game in which voluntary bodily movement is required to control gameplay and constitutes an integral component of task performance. In Kinesthesia, participants use full-body movements captured by a motion sensor to interact with therapeutic games displayed on a conventional screen, without a head-mounted virtual-reality device. Thus, the individual Kinesthesia games meet the operational definition of exergames, while the system as

a whole is more precisely described as an exergaming platform. The study setup comprised a standard laptop running Kinesthesia software, a conventional display or television, a Kinect for Xbox 360 full-body depth sensor, and an unobstructed movement area; sessions were delivered under staff supervision. The conventional-screen, non-immersive architecture was selected to preserve environmental awareness and permit portable deployment without a head-mounted display.

Unlike commercial off-the-shelf exergames originally developed for entertainment and subsequently adapted for rehabilitation, and unlike conventional dual-task protocols in which the cognitive and motor components may remain functionally separable, Kinesthesia was designed from first principles to operationalise simultaneously six neuroplasticity prerequisites: attention, accuracy, adequacy, reinforcement, frequency, and flexibility (Nahum et al., 2013; Chiarenza & Bosch-Bayard, 2025; Bosch-Bayard et al., 2026a). Cognitive and motor demands are structurally integrated within each game: interruption of either component prevents successful task completion. The platform further incorporates centimetre-resolution full-body motion capture, bidirectional adaptive difficulty within and across sessions, immediate visual and auditory performance feedback, individualised progression, and participant-level performance recording. A full description of the game suite, adaptive difficulty algorithm, and session structure is provided in Supplementary Materials S3.

2.4 Proposed neurobiological mechanisms

At the network level, simultaneous sensorimotor and cognitive engagement is expected to promote large-scale functional integration and counteract age-related network segregation through coordinated recruitment of prefrontal–cerebellar and cortico–basal-ganglia loops (Doyon & Benali, 2005; Leisman et al., 2016). Aerobic-enriched sensorimotor activity upregulates hippocampal neurogenesis (Wollesen & Voelcker-Rehage, 2014), providing a substrate for the episodic memory gains that constitute the earliest marker of dementia risk. Bilateral midline-crossing movements — a structural feature of every Kinesthesia game — engage interhemispheric connectivity via the corpus callosum, and the proprioceptive–vestibular richness of full-body interaction may contribute independently to cerebellar plasticity and network synchronisation. At the synaptic level, the adequacy–reinforcement–frequency triad operationalised by the adaptive algorithm is expected to support long-term potentiation and functional network reorganisation (Nahum et al., 2013). All proposed mechanisms are theoretical constructs pending empirical validation; paired resting-state EEG suitable for analysis was available in 19 of 20 participants and neurophysiological findings are reported separately as convergent evidence.

2.5 Outcome measures

Assessments were administered at baseline and within two weeks of the 30th session by trained evaluators blinded to individual game performance data. Primary cognitive outcomes were derived from the NEUROPSI Atención y Memoria battery (Ostrosky-Solís et al., 2007), which provides domain-specific measurement across subscales (orientation, attention, executive function, working

memory, immediate and delayed memory) without ceiling effects, incorporates age- and education-stratified normative data, and has been validated in Spanish-speaking older adults. Primary motor outcome was the Six-Minute Walk Test (6MWT): total metres walked and theoretical percentage of the age-expected normative value (Enright equation; Enright & Sherrill, 1998). The Tinetti Performance-Oriented Mobility Assessment and Timed-Up-and-Go test were administered at baseline for sample characterisation only and are not included as repeated-measures outcomes.

Paired resting-state EEG recordings were acquired using a 19-channel cap (10–20 system, linked-earlobe reference) and were suitable for analysis in 19 of 20 participants. One participant's post-intervention recording was excluded because extensive artefacts rendered it unusable under the study's quality-control procedures. The protocol did not use algorithmic artefact-removal procedures to salvage recordings that failed quality control, and because the problem was identified after project completion, repeat acquisition was not possible. Spectral indices were extracted per standard quantitative EEG protocols and analysed as a secondary neurophysiological characterisation. Adherence (proportion of scheduled sessions completed) and post-intervention satisfaction were collected as primary feasibility outcomes. Satisfaction was assessed after the intervention using an investigator-developed Spanish-language questionnaire rather than a published validated scale. Participants provided 1–5 numeric ratings for seven games or game groupings and 12 items concerning the equipment and service, followed by questions about continued use and suggested improvements. Ratings were summarised descriptively by item; no psychometric total score or cut-off was interpreted (Supplementary Appendix A).

2.6 Statistical analysis

Pre-to-post differences were assessed by sign-flip permutation tests (5,000 permutations), implemented by sign reversal of paired differences under the null and suitable for small paired samples. Multiple comparisons were controlled by Benjamini–Hochberg (BH) false discovery rate correction applied separately within each instrument domain (NEUROPSI: 39 tests; 6MWT: 12 tests). Results with $q < 0.05$ were deemed statistically significant; $q = 0.05–0.10$ were classified as trends; uncorrected $p < 0.05$ but $q \geq 0.05$ are flagged as (*). Effect sizes are Cohen's d (mean delta / SD of deltas). An exploratory MoCA-gradient analysis used permutation-based Spearman rank correlations with BH correction within domain. For game–outcome correlations, within-game learning was represented by signed R^2 ($sR^2 = \text{sign of the regression slope} \times R^2$) and BH correction was applied within each game (Supplementary Table S1). Positive score sR^2 values indicate increasing scores; negative time sR^2 values indicate decreasing completion time and therefore faster performance with practice. All findings are exploratory. NEUROPSI combined $N = 20$; 6MWT combined $N = 19$ (one HC participant lacked a valid 6MWT).

3. Results

3.1 Adherence and feasibility

Adherence was 100% in both subgroups: all participants who reached the post-intervention assessment completed every scheduled session without dropout attributable to the intervention. No adverse events related to Kinesthesia were reported. Satisfaction ratings on the investigator-developed 1–5 questionnaire were uniformly high: MCI participants rated all evaluated games or game groupings 4.25–5.0/5 (mean ≈ 4.75); HC participants rated all evaluated items above 4.5/5, with near-universal willingness to continue the programme. These descriptive findings support high acceptability of Kinesthesia across both cognitive profiles.

3.2 Cognitive outcomes

The primary analysis (N=20, BH correction within 39 NEUROPSI tests) identified BH-significant improvements in eight variables ($q \leq 0.05$): spontaneous verbal recall ($d = +1.07$, $q = 0.004$), cued verbal recall ($d = +0.93$, $q = 0.004$), memory-learning curve ($d = +0.85$, $q = 0.014$), verbal recognition ($d = +0.65$, $q = 0.040$), associated pairs evocation ($d = +0.81$, $q = 0.023$), name evocation ($d = +0.75$, $q = 0.025$), sequential series ($d = +1.07$, $q = 0.010$), and global normalised score ($d = +0.93$, $q = 0.008$). Six additional variables showed BH trends ($q < 0.10$). Full data are presented in Table 1 and Figure 1.

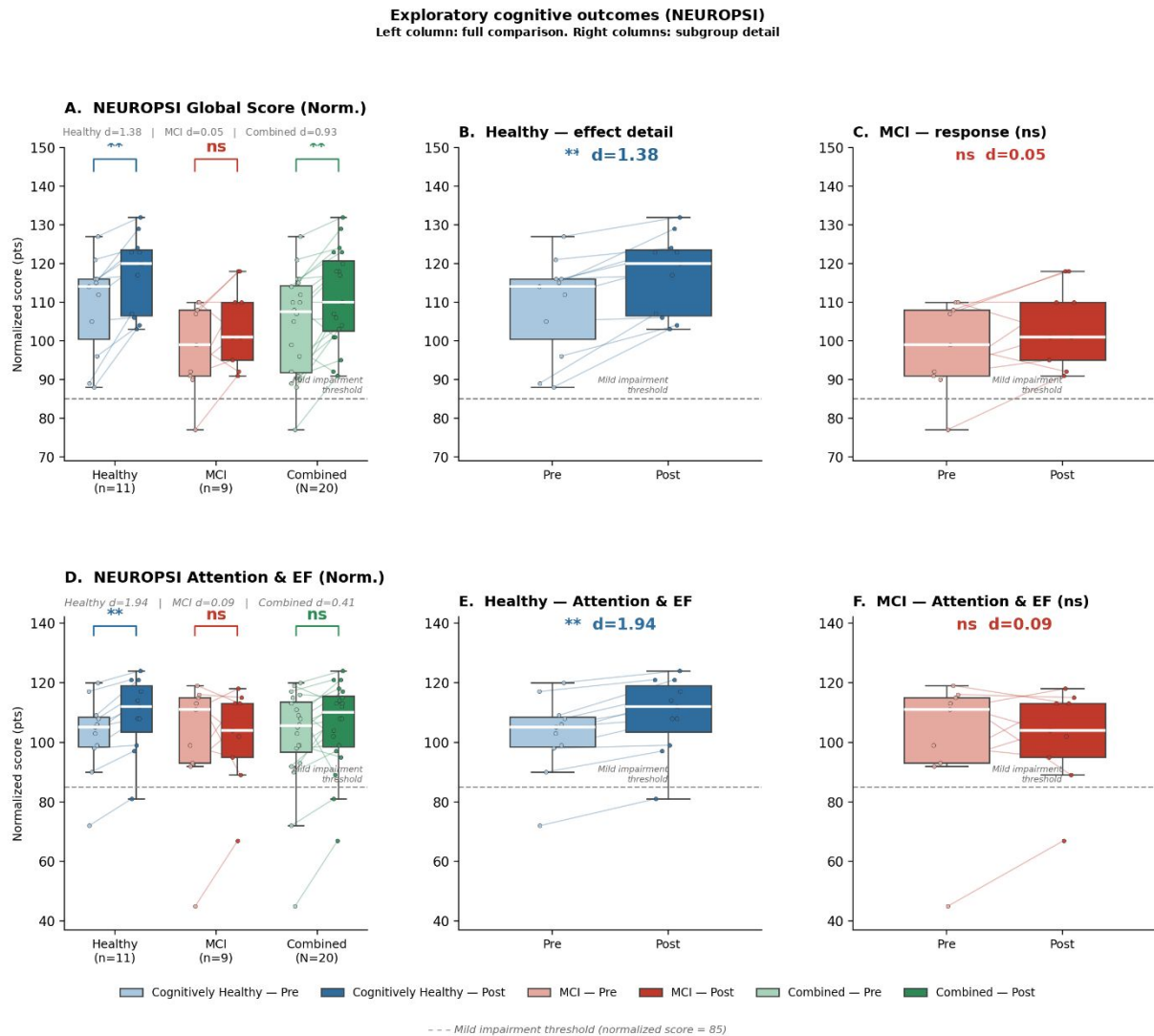


Figure 1. Exploratory NEUROPSI cognitive outcomes (NEUROPSI $N=20$; HC $n=11$, MCI $n=9$). Panels A and D show the full pre-to-post comparison for the global normalised score (A) and the Attention & Executive Function normalised composite (D) across the cognitively healthy (HC), MCI, and combined samples; panels B/C and E/F give the corresponding subgroup detail. Boxplots show pre (light) and post (dark) distributions with individual trajectories overlaid; the dashed line marks the mild-impairment threshold (normalised score = 85). Effect sizes are Cohen's d (Δ/SD of deltas). In HC, both composites improved with large, BH-significant effects (global $d=1.38$, $q=0.012$; Attention/EF $d=1.94$, $q=0.012$); in MCI, neither global score ($d=0.05$, $p=0.867$) nor Attention/EF score ($d=0.09$, $p=0.793$) changed significantly. At the combined level, the global normalised score improved ($d=0.93$, $p<0.001$, BH $q=0.008$), whereas the Attention/EF composite did not ($d=0.41$, $p=0.084$). These within-subgroup results do not establish a differential response because between-group differences in change were not formally tested. All findings are exploratory; the study lacked a randomised control condition.

Alt Text: Six-panel figure showing pre-to-post changes in NEUROPSI cognitive scores. Top row: global normalised score for combined sample (A), cognitively healthy HC subgroup (B), and MCI subgroup (C). Bottom row: Attention and Executive Function composite (D, E, F). Each panel shows boxplots with individual trajectories overlaid; blue represents HC, red represents MCI. A dashed horizontal line marks the mild-impairment threshold at normalised score 85. HC shows large improvements crossing into the high-normal range; MCI shows no significant change.

Exploratory subgroup analyses suggested descriptively different profiles. In the MCI subgroup (n=9), eight variables showed uncorrected significance ($p < 0.05$), concentrating on verbal memory ($d = +1.29$ – 1.67) and executive sequencing ($d = +1.28$). In the HC subgroup (n=11), two variables survived BH correction: normalised Attention/Executive score ($d = +1.94$, $q = 0.012$) and global normalised score ($d = +1.38$, $q = 0.012$). This apparent pattern—uncorrected MCI gains concentrated in impaired domains and BH-significant HC gains in normalised composites—generates a hypothesis consistent with cognitive reserve theory (Khalaila et al., 2024; Liu et al., 2023), but between-group differences in change were not formally tested. The exploratory MoCA-gradient analysis showed no variable surviving BH correction for the MoCA–delta association; therefore, it did not provide corrected evidence that improvement varied with baseline MoCA.

3.3 Motor outcomes

The primary analysis identified BH-significant improvements in both 6MWT variables: metres walked ($d = +1.20$, $p < 0.001$, $q = 0.002$) and theoretical percentage of the age-expected norm ($d = +0.76$, $p = 0.002$, $q = 0.023$). Mean gain was $+60.2$ m (pre: 415.6 ± 75.1 m; post: 475.9 ± 58.5 m), substantially exceeding the 14–30 m range reported as clinically important across adult clinical populations (Bohannon & Crouch, 2017). Subgroup analyses showed consistent direction: MCI $d = +1.40$ ($p = 0.004$, $q = 0.024$) and HC $d = +1.06$ ($p = 0.006$, $q = 0.024$), both BH-significant. The convergence of subgroup direction with the BH-confirmed combined result strengthens confidence in this finding (Figure 2).

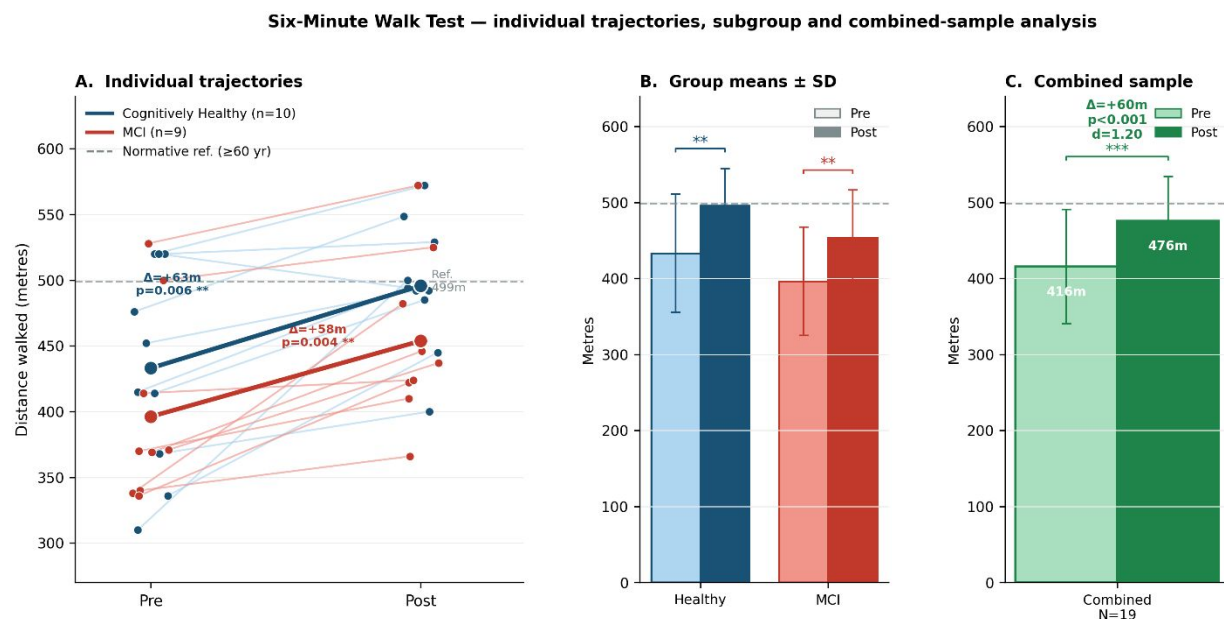


Figure 2. Six-Minute Walk Test—individual trajectories, subgroup, and combined-sample analysis (6MWT N=19; HC n=10, MCI n=9). (A) Individual pre-to-post trajectories (thin lines) with subgroup means (thick lines); blue = HC, red = MCI; the dashed line marks the age-expected normative reference (Enright equation). (B) Subgroup means $\pm$ SD, pre versus post. (C) Combined-sample means. Distance walked increased by $+62.6$ m in HC ($d = 1.06$, $p = 0.006$) and $+57.6$ m in MCI ($d = 1.40$, $p = 0.004$); the combined gain was $+60.2$ m ($d = 1.20$, $p < 0.001$, BH $q = 0.002$), exceeding

the 14–30 m range reported as clinically important across adult clinical populations (Bohannon & Crouch, 2017). All findings are exploratory; the study lacked a randomised control condition.

Alt Text: Three-panel figure showing Six-Minute Walk Test results. Panel A shows individual pre-to-post trajectories as thin lines with thick subgroup mean lines; blue for cognitively healthy HC, red for MCI; a dashed line marks the age-expected normative reference. Panel B shows subgroup means plus or minus SD pre versus post. Panel C shows combined-sample means. Both groups improved substantially, with combined gain of 60.2 metres above the minimally important difference threshold.

3.4 Neurophysiological correlates

Of 10 pre-specified MoCA-gradient spectral EEG indices, 8 survived BH correction ($q < 0.05$) at baseline, confirming that pre-intervention neurophysiological state covaried with the cognitive reserve continuum. Spectral indices were evaluated against a multinational normative database spanning nine countries and approximately 2,000 participants, allowing individual deviations from age-expected norms to be quantified. Post-intervention, 21 of 23 pre-existing spectral differences between subgroups resolved, while 44 new spectral patterns emerged — a reorganisation characterised by reduction of slow-wave excess and emergence of alpha-range activity, consistent with spectral normalisation accompanying the observed behavioural improvements. Full EEG methodology and spectral index results are reported separately (Bosch-Bayard et al., 2026b); neurophysiological findings are summarised here as convergent evidence only.

3.5 Within-platform learning and game–outcome associations

Session-by-session performance data demonstrated systematic skill acquisition across all analysed games: mean regression slopes for scores were positive and for execution times were negative, with 80–91% of participants showing individually significant improvement trajectories in the highest-exposure games (Categorization variants, Follow the Trail, Two Steps, and Lit Tiles; $p < 0.001$). To preserve learning direction, the plotted index is signed R^2 ($sR^2 = \text{sign of slope} \times R^2$): positive score sR^2 denotes increasing scores, whereas negative time sR^2 denotes decreasing completion time and thus faster performance with practice. Figure 3 shows six selected BH-significant associations. In the combined sample, stronger Categorized N time learning was associated with greater raw-memory gain ($\rho = +0.79$, $q = 0.003$, $n = 17$), and stronger Object Categorization time learning was associated with greater gain in 6MWT metres ($\rho = -0.61$, $q = 0.015$, $n = 18$). In HC, time learning in Follow the Trail and Lit Tiles was associated with 6MWT gains ($\rho = -0.76$, $q = 0.015$, $n = 10$ and $\rho = -0.74$, $q = 0.020$, $n = 10$, respectively); because improvement in time is encoded by increasingly negative sR^2 , these negative correlations indicate that larger practice-related time reductions accompanied larger positive walking gains. In MCI, Categorized N score learning correlated positively with Stroop correct-response gains ($\rho = +0.93$, $q = 0.038$, $n = 7$), whereas Follow the Trail score learning correlated negatively with category-formation change ($\rho = -0.93$, $q = 0.037$, $n = 7$). These exploratory associations do not establish causality, and the small, outcome-specific pairwise samples limit interpretation.

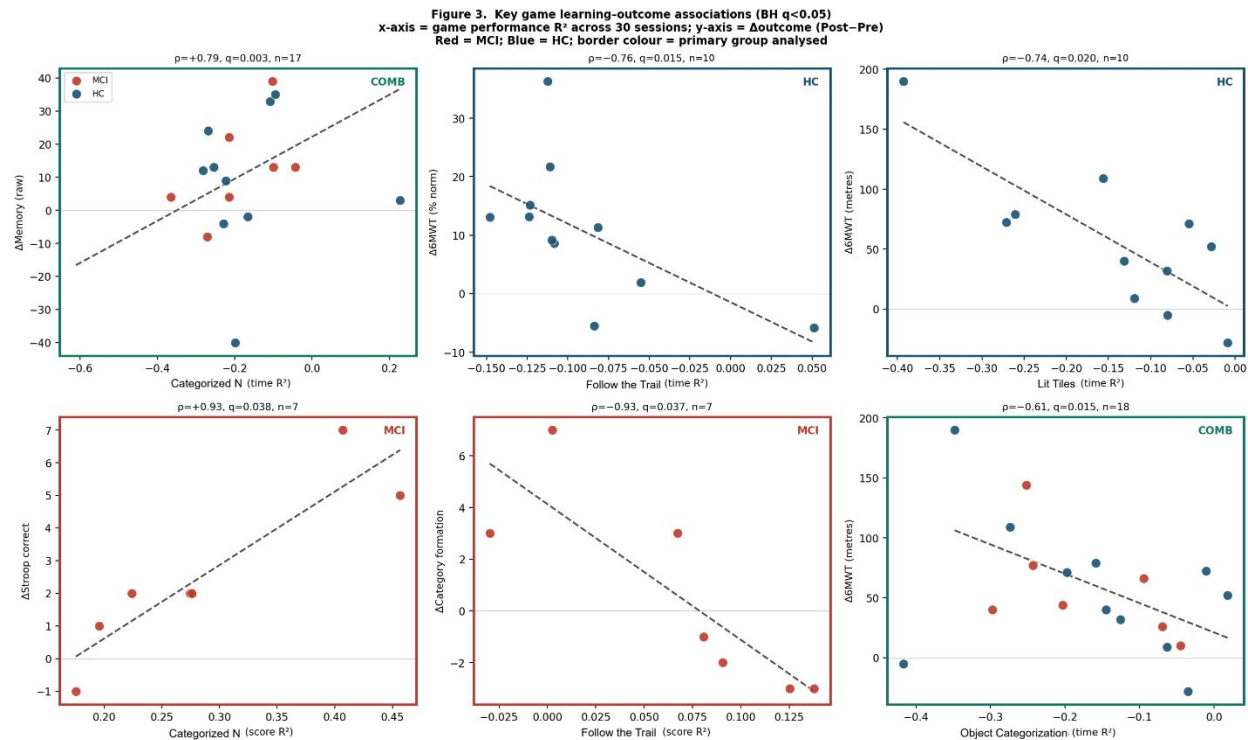


Figure 3. Selected game learning–outcome associations surviving Benjamini–Hochberg FDR correction ($q < 0.05$). Each panel shows the Spearman correlation between signed within-game learning (x-axis: $sR^2 = \text{sign of the regression slope} \times R^2$ across 30 sessions) and change in a clinical outcome (y-axis: Post–Pre). For score measures, positive sR^2 indicates increasing scores. For completion-time measures, negative sR^2 indicates decreasing time—that is, faster performance with practice—so a negative correlation with a positive outcome change can represent concordant improvement. Red dots denote MCI and blue dots HC; the coloured frame identifies the analysed group (green = combined, red = MCI, blue = HC). Dashed lines are linear fits for the analysed group; Spearman ρ , BH-corrected q , and pairwise n are printed above each panel. Top row: Categorized N time sR^2 versus raw-memory change in the combined sample ($\rho = +0.79$, $q = 0.003$, $n = 17$); Follow the Trail time sR^2 versus 6MWT percentage change in HC ($\rho = -0.76$, $q = 0.015$, $n = 10$); and Lit Tiles time sR^2 versus 6MWT metres change in HC ($\rho = -0.74$, $q = 0.020$, $n = 10$). Bottom row: Categorized N score sR^2 versus Stroop correct-response change in MCI ($\rho = +0.93$, $q = 0.038$, $n = 7$); Follow the Trail score sR^2 versus category-formation change in MCI ($\rho = -0.93$, $q = 0.037$, $n = 7$); and Object Categorization time sR^2 versus 6MWT metres change in the combined sample ($\rho = -0.61$, $q = 0.015$, $n = 18$). All associations are exploratory.

Alt Text: Six-panel scatter-plot figure of selected BH-significant game learning–outcome associations. The x-axis is signed R -squared for each participant's performance trajectory across 30 sessions; negative values on time-based panels mean completion time decreased with practice, while positive values on score-based panels mean scores increased. The y-axis is clinical change from pre to post. Panels show associations for Categorized N, Follow the Trail, Lit Tiles, and Object Categorization with memory, Stroop, category formation, or Six-Minute Walk Test change. Red points are MCI and blue points HC; frame colour identifies the analysed group.

Table 1. Pre-to-post cognitive and functional outcomes: effect sizes and Benjamini–Hochberg FDR correction.

Outcome	MCI (n=9)				HC (n=11)				Combined (N=20)						
	d	p	q		d	p	q		Pre M±SD	Post M±SD	Δ	d	p	q	
NEUROPSI — Executive functions															
Sequential series	+1.28	0.029	0.164	(*)	+0.89	0.032	0.182	(*)	1.58±1.22	2.47±0.94	+0.89	+1.07	0.001	0.010	**
Stroop correct	+0.98	0.029	0.164	(*)	+0.28	0.424	0.688		—	—	—	+0.57	0.030	0.089	~
Categorization	+0.23	0.609	0.911		+1.06	0.011	0.107	(*)	—	—	—	+0.63	0.015	0.053	~
Norm. Attn/Exec.	+0.09	0.793	0.967		+1.94	<0.001	0.012	**	—	—	—	+0.41	0.084	0.204	
Norm. global score	+0.05	0.867	—		+1.38	0.002	0.012	**	105.4±13.9	110.0±12.9	+4.6	+0.93	<0.001	0.008	**
NEUROPSI — Verbal memory															
Memory-learning curve	+1.67	0.007	0.137	(*)	+0.49	0.197	0.559		—	—	—	+0.85	0.002	0.014	**
Spont. verbal recall	+1.38	0.015	0.137	(*)	+0.85	0.033	0.182	(*)	6.63±2.75	8.26±2.22	+1.63	+1.07	<0.001	0.004	**
Cued verbal recall	+1.29	0.016	0.137	(*)	+0.69	0.073	0.315		—	—	—	+0.93	<0.001	0.004	**
Verbal recognition	+0.88	0.036	0.174	(*)	+0.53	0.171	0.559		—	—	—	+0.65	0.008	0.040	**
Assoc. pairs evoc.	+0.58	0.175	0.568		+1.07	0.015	0.120	(*)	8.16±2.9	9.74±2.4	+1.58	+0.81	0.004	0.023	**
Name evocation	+0.89	0.066	0.286		+0.69	0.071	0.315		—	—	—	+0.75	0.004	0.025	**
6-Minute Walk Test (N=19; HC n=10, MCI n=9)															
6MWT metres (m)	+1.40	0.004	0.024	**	+1.06	0.006	0.024	**	415.6±75.1	475.9±58.5	+60.2	+1.20	<0.001	0.002	**
6MWT % theoretical	+0.48	0.129	—		+1.11	0.006	0.024	**	77.6±13.9	87.8±12.4	+10.2	+0.76	0.002	0.023	**

d = Cohen's d (mean delta/SD of deltas); p = sign-flip permutation p-value (5,000 permutations); q = BH-corrected p-value. BH correction applied separately within NEUROPSI (39 tests) and 6MWT (12 tests). Flags: ** q<0.05; ~ q<0.10 trend; (*) p<0.05 uncorrected only. Full pre-to-post means ± SD in Supplementary Table S2.

Table 2. Alignment of Kinesthesia with six neuroplasticity-based rehabilitation principles (Nahum et al., 2013) and proposed neurophysiological monitoring markers.

BP = Bereitschaftspotential; MCP = Motor Cortex Potential; SPP = Skilled Performance Positivity; PAN = Post-Action Negativity; MRP = Movement-Related Potential. For a review of MRP applications in cognitive rehabilitation monitoring, see (Chiarenza & Bosch-Bayard, 2025).

Principle	Definition	Implementation in Kinesthesia	Proposed MRP marker
Attention	Training must engage sustained, focused attention; passive engagement is insufficient for plastic change	Simultaneous cognitive and motor demands require undivided attention throughout each task; attentional disengagement produces immediate performance failure, providing intrinsic motivation to sustain engagement	<i>BP amplitude; N100 (sensory attention gating)</i>
Accuracy	Responses must be spatially and temporally precise; imprecision must be detected and corrected	Full-body motion capture (centimetre resolution) enforces precise responses in every game; only accurate motor outputs are reinforced by the game narrative	<i>MCP (sensorimotor integration); P200 (motor completion)</i>
Adequacy	Difficulty must match individual capacity — neither too easy nor too demanding — and adapt continuously	Bidirectional adaptive algorithm adjusts stimulus speed, density, and spatial range within each game in real time; difficulty level is retained across sessions for each participant	<i>BP onset and amplitude (strategic motor programming)</i>
Reinforcement	Successful performance must be immediately and distinctively rewarded to gate synaptic strengthening	Real-time visual and auditory feedback on every correct response; emotionally engaging game narrative provides immediate reward; structured feedback distinguishes correct from incorrect responses	<i>SPP (performance outcome evaluation); PAN (error detection)</i>
Frequency	Sufficient repetition density and session frequency are required for consolidation	30 sessions × 60 minutes (≈30 min active play per session), 3×/week, 10 weeks; within-session repetition density and between-session spacing optimise consolidation	<i>BP amplitude reduction across sessions (motor engram formation)</i>
Flexibility	Programme must be individually adaptable and generalisable across settings	Modular 13-game study suite; individualised session composition; applicable from normal aging to MCI; portable architecture designed for clinical, community, home-based, and low-resource deployment. Real-world implementation and scaling outcomes require direct evaluation.	<i>SPP latency trajectory across sessions</i>

4. Discussion

This uncontrolled pilot provides preliminary feasibility evidence and outcome signals for Kinesthesia in community-dwelling older adults spanning the healthy to MCI spectrum. The primary findings—BH-corrected improvements in eight cognitive variables spanning verbal memory, executive sequencing, and global performance ($d=0.65$ – 1.07) and both 6MWT outcomes ($d=0.76$ – 1.20 ; $N=19$), alongside cognitive analyses in $N=20$, together with 100% adherence and high satisfaction—constitute the core result. The neurophysiological findings provide convergent exploratory evidence: post-intervention resolution of 21 of 23 pre-existing EEG spectral

differences is consistent with, but does not establish, neural reorganisation accompanying behavioural improvement. All findings are exploratory; causal attribution to the intervention requires a randomised controlled design.

Published meta-analytic benchmarks report effect sizes of $g=0.21-0.37$ on global cognition for combined cognitive–physical programmes in MCI (Karssemeijer et al., 2017; Wollesen & Voelcker-Rehage, 2014) and $d=0.3-0.6$ for exergame-based approaches (Gallou-Guyot et al., 2020). The present BH-confirmed effect sizes ($d=0.65-1.94$) exceed these benchmarks, though uncontrolled pre-post designs systematically overestimate effects relative to randomised trials, and practice effects on neuropsychological instruments constitute a plausible contributor in the absence of a control condition. The 6MWT gain (+60.2 m across 10 weeks) substantially exceeds the 14–30 m range reported as clinically important across adult clinical populations (Bohannon & Crouch, 2017), suggesting potential clinical relevance for functional aerobic capacity and mobility in older adults.

A theoretically informative secondary observation was the descriptive subgroup pattern; it should not be interpreted as a formally demonstrated group $\times$ time interaction. The MCI subgroup showed its largest uncorrected gains in verbal memory ($d=+1.29-1.67$) and executive sequencing ($d=+1.28$), whereas the HC subgroup showed BH-significant gains in the executive composite ($d=+1.94$) and global normalised performance ($d=+1.38$). These profiles suggest, rather than establish, a cognitive-reserve-dependent hypothesis (Khalaila et al., 2024; Liu et al., 2023). The terms ‘repair’ and ‘expansion’ are used here as heuristic interpretations: improvement in MCI could reflect reinforcement of partially degraded networks, whereas improvement in HC could reflect recruitment of additional circuits. The post-intervention EEG pattern—resolution of 21 of 23 pre-existing between-subgroup spectral differences alongside emergence of 44 new patterns—is compatible with that interpretation, but individual-level EEG trajectory analysis and a controlled design are required to test it. The Scaffolding Theory of Aging and Cognition (Park & Reuter-Lorenz, 2009) offers complementary framing for how adaptive task difficulty might engage different compensatory processes according to baseline capacity. Accordingly, the present subgroup pattern should be treated as a hypothesis for stratified evaluation in a future randomised trial, not as evidence that Kinesthesia has established restorative and performance-enhancing mechanisms in different cognitive groups.

The resting-state EEG findings provide convergent neurophysiological evidence: the majority of pre-specified spectral indices covaried with the cognitive reserve continuum at baseline, consistent with prior evidence that quantitative EEG tracks MCI-related neurophysiological change (Babiloni et al., 2009; Smailovic & Jelic, 2019). Evaluated against a multinational normative reference spanning nine countries, the post-intervention spectral reorganisation — characterised by reduction of slow-wave excess and emergence of alpha-range activity in regions previously showing pathological deviation — is consistent with spectral normalisation accompanying behavioural improvement. This pattern aligns with the neuroplastic mechanisms hypothesised to

underlie sensory-motor cognitive training (Chiarenza, 1982; Ali et al., 2022), though causal attribution requires a controlled design with an EEG comparator condition.

The domain-specific game–outcome correlations — locomotor games predicting walking capacity in HC; attentional and sequential ordering games predicting cognitive gains in MCI — suggest that individual game mechanics may engage and reinforce task-congruent neural circuits, consistent with scaffolding theory (Park & Reuter-Lorenz, 2009). However, with $n=7-11$ per subgroup these correlations are exploratory and may reflect shared baseline capacity rather than specific transfer.

4.1 Exergame type, immersion level, and clinical implementation

Development purpose and immersion level are independent dimensions: both commercial off-the-shelf and purpose-built therapeutic exergames may be presented through non-immersive or fully immersive interfaces. Kinesthesia occupies the purpose-built, non-immersive category. Its games were designed around predefined cognitive-motor targets and neuroplasticity prerequisites rather than repurposed from entertainment software, while participants remain visually and physically anchored in the real environment and control gameplay through full-body movement on a conventional screen. The domain-specific game-outcome associations observed here are compatible with that therapeutic design rationale, but this uncontrolled study cannot determine whether purpose-built content is superior to adapted commercial games. Whether a fully immersive implementation would produce similar or greater effects also remains unknown. Comparative evidence does not support a simple dose-response relation between technological immersion and cognitive benefit. A Bayesian network meta-analysis of 51 randomised trials in older adults found the most consistent benefits for global cognition and executive function with non-immersive exergames, although immersive VR ranked highly for global cognition and the certainty of most comparisons was very low (Hu et al., 2026). In older adults with MCI, another network meta-analysis ranked semi-immersive and non-immersive approaches above fully immersive VR for global cognition (Qin et al., 2025). Conversely, a small direct randomised comparison of immersive and non-immersive versions of the same exergame reported greater gains in selected executive outcomes with immersion, partly mediated by perceived presence (Huang, 2020). Thus, immersion may enhance presence and engagement for some outcomes, whereas non-immersive systems preserve environmental awareness and avoid head-mounted-display requirements. Direct head-to-head comparisons with adapted commercial exergames and fully immersive therapeutic platforms would be required before claiming equivalence or superiority for Kinesthesia.

4.2 Limitations

The principal limitation is the absence of a randomised control group: observed pre-to-post changes cannot be causally attributed to the Kinesthesia intervention. Practice effects on repeated neuropsychological administration, regression to the mean, therapist attention, and natural fluctuation in functional capacity may contribute substantially. The small sample ($N=20$; $n=9$ MCI,

n=11 HC) limits statistical power: with n=9-11, the sign-flip test achieves approximately 55% power for d=0.80 at alpha=0.05 with BH correction across 39 tests. The sample was recruited from a single community centre with marked gender imbalance (85% women in the analysed sample); generalisability cannot be assumed. The substantial educational difference between subgroups (approximately 6.3 years, p<0.001) is a potential confound for cognitive outcomes not fully resolved by MoCA stratification alone. The HC and MCI labels were operational classifications based on MoCA screening rather than formal diagnoses; because one participant scored 15 on the MoCA, the MCI subgroup may have included more substantial impairment. This classification does not meet NIA-AA or DSM-5 diagnostic criteria and limits subgroup interpretation. Finally, the gap between data collection (2017) and publication reflects institutional and resource constraints rather than methodological concerns; the study protocol and analytical approach are consistent with current standards. Continued use of the discontinued Kinect for Xbox 360 is a deliberate cost-conscious engineering decision, not a consequence of limited technical capacity. A developer price check in September 2026 found replacement Kinect 360 units on Amazon, eBay, Mercado Libre, and other online marketplaces for under US\$50, whereas technically suitable Intel RealSense depth-camera alternatives were listed from approximately US\$400, with many configurations priced substantially higher. This dated price snapshot—not a formal market or cost-effectiveness analysis—documents the principal reason for retaining the Kinect 360 and the non-immersive architecture. In more than ten years of platform use, none of the Kinect 360 units used by the team has failed in service; its three-dimensional skeletal tracking and mature SDK meet all current game requirements. Newer cameras offer greater precision and sampling rates, but those capabilities exceed the platform's present requirements and would not provide a corresponding functional advantage. Long-term secondary-market availability cannot be guaranteed. Because hardware-specific communication is isolated in a single module, migration would require adapting and validating that module rather than redesigning the therapeutic games.

4.3 Implications for a confirmatory trial

The BH-confirmed effect sizes provide preliminary inputs for RCT planning. For 80% power at $\alpha=0.05$ in a two-sided two-group comparison, detecting d=0.80 requires approximately 26 participants per arm and d=1.00 approximately 17 per arm. Allowing 20% attrition gives recruitment targets of approximately 33 and 22 participants per arm, respectively. Because effect sizes from an uncontrolled pre-post pilot may be inflated, these figures are optimistic planning estimates; a definitive calculation should be based on the expected between-group effect and the prespecified primary analysis. The confirmatory RCT should incorporate an active control condition (e.g., conventional dual-task walking programme or social exercise group), stratified randomisation by MoCA and educational level, pre-registered primary outcomes (delayed verbal memory composite; 6MWT metres walked), assessor blinding, and follow-up at 3 and 6 months to evaluate durability of gains. Beyond the confirmatory RCT, a longer-term research agenda encompasses validation of remote supervision and home-based delivery modalities; extension to other populations sharing overlapping cortico-striato-cerebellar targets; and evaluation in low- and

middle-income settings, where implementation feasibility, resource use, accessibility, and clinical outcomes require direct study.

5. Conclusions

Kinesthesia demonstrated strong feasibility—100% adherence and uniformly high satisfaction across both cognitive profiles—and large-effect improvement signals in verbal memory, global cognitive function, and functional walking capacity in a small pilot sample of community-dwelling older adults with and without MCI. Convergent neurophysiological findings support the behavioural signal. The descriptive MCI–HC pattern is a hypothesis for confirmatory evaluation; between-group differences in change were not formally tested. All findings are exploratory, and proposed mechanisms remain theoretical pending replication in a randomised controlled trial with active comparators. Kinesthesia's modular software, conventional display, compact depth sensor, and absence of a head-mounted device make portability, accessibility, and deployment across clinical, community, home-based, and resource-limited settings explicit engineering characteristics of the platform. The extent to which those characteristics translate into lower total implementation cost, reduced staff or training burden, and successful real-world scaling remains to be established in implementation research.

CRedit authorship contribution statement

Jorge Bosch-Bayard: Conceptualization, Methodology, Formal analysis, Software, Visualization, Writing – original draft, Writing – review & editing, Supervision, Project administration. **Judith Guerrero-Sauzamedia:** Investigation, Data curation, Resources. **Laura Navarrete-Torres:** Investigation, Data curation, Resources. **Rodolfo Bosch-Bayard:** Conceptualization, Methodology, Investigation, Resources, Validation, Writing – review & editing. **Giuseppe A. Chiarenza:** Conceptualization, Validation, Writing – original draft, Writing – review & editing. **Liane Aguilar-Fabre:** Conceptualization, Methodology, Formal analysis, Supervision, Validation, Writing – review & editing. **René Rodríguez-Valdés:** Conceptualization, Methodology, Formal analysis, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Ethics approval

This study was approved by the Research Ethics Committee of the Universidad Autónoma de Querétaro (protocols 02-03/11/2016 and 03-01/12/2016) and was conducted in accordance with the Declaration of Helsinki.

Informed consent

All participants provided written informed consent prior to enrolment.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used Claude (Anthropic) and ChatGPT/Codex (OpenAI) to assist with language editing, consistency checking, document formatting, and improvement of the clarity and readability of the manuscript text in English. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Kinesthesia — Supplementary Material *Bosch-Bayard et al., 2026*

Kinesthesia: feasibility and preliminary efficacy of a sensory-motor cognitive intervention in community-dwelling older adults with and without Mild Cognitive Impairment

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Supplementary Table S1. Selected game–outcome Spearman correlations

Spearman ρ between signed game-learning indices ($sR^2 = \text{sign of slope} \times R^2$) and pre-to-post clinical change, by subgroup. Positive score sR^2 indicates increasing scores; negative time sR^2 indicates decreasing completion time (faster performance with practice). BH FDR correction was applied within each game and subgroup across the 11 outcomes presented in Table S1.

Flags: ** $q < 0.05$ (BH-significant); ~ $q < 0.10$ (trend); — $n < 4$ (insufficient data). Blue shading: BH-significant. Green: trend. Row shading (blue): combined-sample result. All analyses exploratory.

Game	Grp	n	NEUROPSI outcomes (Spearman ρ)								6MWT		
			Mem curve	Verb spont	Verb cued	Verb recog	Assoc pairs	Name evoc	Stroop	Categ	Memory (raw)	6MWT (m)	6MWT (%)
Locomotor games													
Two Steps	MCI	7	−0.46	−0.80	−0.41	+0.38	+0.39	−0.56	+0.71	+0.68	−0.38	+0.43	+0.43
	HC	11	—	+0.27	−0.18	−0.68	−0.16	+0.21	+0.24	+0.11	−0.30	+0.31	−0.41
	All	18	+0.37	+0.37	+0.64~	−0.05	+0.37	−0.33	+0.11	−0.05	−0.28	+0.25	+0.15
Follow the Trail	MCI	7	+0.15	+0.71	+0.71	−0.47	−0.29	−0.09	−0.67	−0.93**	−0.86~	+0.11	−0.25
	HC	11	+0.72	+0.72	+0.20	+0.40	−0.40	+0.57	+0.33	+0.19	−0.14	−0.47	−0.76**
	All	18	−0.17	−0.17	+0.35	−0.20	−0.35	+0.20	+0.18	−0.37	−0.41	−0.10	−0.26
Lit Tiles	MCI	7	+0.66	+0.66	−0.60	+0.14	+0.25	−0.72	+0.30	+0.58	+0.21	+0.75	−0.32
	HC	11	−0.47	−0.47	−0.41	+0.29	−0.26	+0.68	+0.46	−0.33	+0.28	−0.74**	+0.13
	All	18	−0.15	−0.15	−0.40	−0.06	+0.16	+0.10	+0.23	+0.21	+0.22	−0.17	+0.05

<i>Attentional search</i>													
Categorization	MCI	7	-0.58	-0.58	-0.40	+0.29	-0.55	+0.75	+0.11	+0.29	+0.18	-0.73	-0.61
	HC	11	+0.22	+0.22	+0.58	+0.33	+0.25	-0.16	-0.60	-0.32	+0.25	+0.41	+0.44
	All	18	-0.29	-0.29	-0.17	-0.05	-0.40	-0.26	+0.29	+0.43	+0.22	-0.61**	-0.37
Categorized N	MCI	7	+0.31	+0.31	+0.82	-0.79	-0.33	-0.04	+0.93**	-0.51	+0.61	+0.51	-0.68
	HC	11	-0.26	-0.26	-0.03	+0.23	-0.14	+0.61	-0.58	+0.24	-0.38	-0.04	+0.14
	All	17	+0.40	+0.40	+0.20	-0.13	-0.08	-0.11	+0.36	-0.21	+0.79**	+0.07	-0.29
Categorized 6	MCI	7	-0.46	-0.46	+0.56	-0.20	-0.33	+0.53	-0.33	-0.78	-0.46	-0.78	+0.71
	HC	11	-0.29	-0.29	+0.32	-0.39	+0.18	-0.22	-0.28	-0.23	+0.18	-0.32	-0.23
	All	18	+0.36	+0.36	+0.59	+0.21	+0.39	+0.39	-0.09	-0.37	-0.21	+0.29	+0.20
<i>Sequential ordering</i>													
Ascending Sort	MCI	7	-0.93**	-0.93**	-0.20	+0.58	+0.35	-0.75	+0.41	-0.13	+0.07	+0.84	+0.36
	HC	11	+0.19	+0.19	+0.25	-0.23	+0.29	+0.48	-0.06	-0.24	-0.12	+0.21	-0.52
	All	18	-0.28	-0.28	+0.16	-0.22	+0.08	+0.14	-0.53	+0.04	+0.04	+0.15	-0.23
<i>Arithmetic</i>													
Math Level 1	HC	8	-0.29	-0.29	+0.53	+0.60	-0.44	+0.27	-0.06	-0.91**	+0.17	-0.49	-0.43
Math Level 3	HC	7	+0.15	+0.15	+0.33	+0.03	+0.19	-0.06	+0.18	-0.94**	+0.50	-0.49	+0.66

Supplementary Table S2. Selected pre-to-post descriptive outcomes aligned with Table 1. Selected pre-to-post means $\pm$ SD, sign-flip permutation p-values, and paired Cohen's d are reported by subgroup and combined sample. Outcome-specific denominators are NEUROPSI N=20 (HC n=11, MCI n=9) and 6MWT N=19 (HC n=10, MCI n=9). The outcomes and numerical values are identical to those in Table 1 of the main manuscript.

Flags reproduce Table 1 of the main manuscript: ** q<0.05; ~ q<0.10 trend; (*) uncorrected p<0.05 with q $\geq$ 0.05; ns = not significant. Effect size d = mean delta / SD of paired deltas. All p-values are from the sign-flip permutation analysis described in the main manuscript; q-values and BH interpretation are reported in Table 1. Blue shading identifies combined-sample BH-significant results or trends.

Outcome measure	Group	Pre M $\pm$ SD	Post M $\pm$ SD	Δ	p	Sig	d	Effect
<i>6-Minute Walk Test</i>								
6MWT Metres (m)	HC (n=10)	433.1 $\pm$ 77.7	495.8 $\pm$ 49.0	+62.6	0.006	**	1.06	LARGE
	MCI (n=9)	396.2 $\pm$ 71.3	453.8 $\pm$ 63.0	+57.6	0.004	**	1.40	LARGE
6MWT % Theoretical	Combined (N=19)	415.6 $\pm$ 75.1	475.9 $\pm$ 58.5	+60.2	<0.001	**	1.20	LARGE
	HC (n=10)	78.4 $\pm$ 15.0	90.9 $\pm$ 8.6	+12.5	0.006	**	1.11	LARGE
	MCI (n=9)	76.7 $\pm$ 13.5	84.3 $\pm$ 15.4	+7.6	0.129	ns	0.48	MED
	Combined (N=19)	77.6 $\pm$ 13.9	87.8 $\pm$ 12.4	+10.2	0.002	**	0.76	LARGE
<i>NEUROPSI — Global scores</i>								
Global Norm. Score	HC (n=11)	108.3 $\pm$ 13.4	116.5 $\pm$ 10.8	+8.2	0.002	**	1.38	LARGE
	MCI (n=9)	102.2 $\pm$ 14.5	102.8 $\pm$ 11.5	+0.6	0.867	ns	0.05	NEGL
	Combined (N=20)	105.4 $\pm$ 13.9	110.0 $\pm$ 12.9	+4.6	<0.001	**	0.93	LARGE
Attn/EF Norm. Score	HC (n=11)	102.1 $\pm$ 13.8	109.0 $\pm$ 13.5	+6.9	<0.001	**	1.94	LARGE

	MCI (n=9)	104.2±14.4	103.7±10.0	-0.6	0.793	ns	0.09	NEGL
	Combined (N=20)	103.1±13.7	106.5±12.0	+3.4	0.084	ns	0.41	SMALL
NEUROPSI — Verbal memory								
Spont. Verbal Recall	HC (n=11)	8.50±2.17	9.80±1.48	+1.30	0.033	(*)	0.85	LARGE
	MCI (n=9)	4.56±1.88	6.56±1.59	+2.00	0.015	(*)	1.38	LARGE
	Combined (N=20)	6.63±2.75	8.26±2.22	+1.63	<0.001	**	1.07	LARGE
Assoc. Pairs Evoc.	HC (n=11)	10.1±1.9	10.9±1.1	+0.80	0.015	(*)	1.07	LARGE
	MCI (n=9)	6.00±2.3	8.44±2.9	+2.44	0.175	ns	0.58	MED
	Combined (N=20)	8.16±2.9	9.74±2.4	+1.58	0.004	**	0.81	LARGE
Cued Verbal Recall	HC (n=11)	8.00±1.83	9.00±1.63	+1.00	0.073	ns	0.69	MED
	MCI (n=9)	5.44±1.33	6.33±1.73	+0.89	0.016	(*)	1.29	LARGE
	Combined (N=20)	6.79±1.99	7.74±2.09	+0.95	<0.001	**	0.93	LARGE
NEUROPSI — Attention & Executive Function								
Successive Series	HC (n=11)	1.60±1.35	2.60±0.84	+1.00	0.032	(*)	0.89	LARGE
	MCI (n=9)	1.56±1.13	2.33±1.00	+0.78	0.029	(*)	1.28	LARGE
	Combined (N=20)	1.58±1.22	2.47±0.94	+0.89	0.001	**	1.07	LARGE
Category Formation	HC (n=11)	14.2±4.4	17.1±4.6	+2.90	0.011	(*)	1.06	LARGE
	MCI (n=9)	10.0±2.8	10.8±4.1	+0.78	0.609	ns	0.23	SMALL
	Combined (N=20)	12.2±4.2	14.1±5.4	+1.90	0.015	~	0.63	MED

Supplementary Materials S3. The Kinesthesia Platform

S3. Kinesthesia Game Suite

Table S3a presents the complete Kinesthesia game suite evaluated in the 2017 study with targeted cognitive domains, specific motor demands, and primary dual-task configuration for each game. Each of the 13 study games is an exergame because voluntary full-body movement is required to control gameplay and is integral to successful task completion. Motor demands common to the suite - spatial displacement, dynamic postural balance, bilateral limb coordination, and body-midline crossing - are embedded across the intervention. Games are ordered from lowest to highest cognitive-motor complexity as applied in the Kinesthesia protocol progression. The adaptive difficulty algorithm adjusts parameters such as stimulus speed, spatial range, response windows, and stimulus density in real time within each game; the ordering shown reflects cross-session progression from foundational to complex integration games.

Abbreviations: EF = Executive Functions; ATT = Attention & Concentration; MEM = Memory; VSP = Visuospatial Processing; CALC = Numerical Calculation; CAT = Categorization; LAT = Laterality; WM = Working Memory.

Game	Cognitive Domains	Specific Motor Demands	Primary Dual-Task Type
Square Hunt	ATT (selective, sustained), EF (response inhibition), VSP	Rapid whole-arm reaching to precise spatial targets; stepping; rapid direction changes	Simultaneous visuomotor tracking under attentional load
Star Rain	ATT (divided, sustained), EF (action planning), VSP	Continuous bilateral upper-limb interception; lateral stepping; arm-foot coordination	Divided attention with continuous movement
The Skier	ATT (sustained), EF (anticipation, trajectory planning), VSP	Whole-body lateral displacement; anticipatory postural adjustments; dynamic balance	Spatial navigation under sustained attentional load
Mathematical Calculation	ATT, CALC, MEM (WM), VSP	Targeted arm reaching; foot placement to correct positions; postural control	High cognitive load simultaneous with precise motor response
3D Geometry	ATT, EF (abstract reasoning, mental rotation), MEM, VSP	Precise arm positioning; body orientation changes; midline crossing	Visuospatial abstraction simultaneous with motor precision
Maze Navigation	ATT, EF (spatial planning, cognitive flexibility), VSP	Continuous whole-arm reaching along maze walls; midline crossing; trunk rotation	Strongest interhemispheric demand in suite
Categorization Games	ATT, EF (cognitive flexibility, rule-based classification), CAT, VSP	Whole-body movement to designated spatial categories; stepping; response inhibition	Simultaneous semantic decision-making and whole-body motor response
Numerical Sequencing (hands)	MEM (sequential WM), EF (planning, sequencing), ATT, VSP	Sequential bilateral hand reaching in memorized order; midline crossing	Sequential WM load simultaneous with motor sequencing
Lit Tiles	ATT (sustained, sequential), EF (sequencing), LAT, VSP	Foot placement on illuminated tiles in sequence; laterality stepping; trunk rotation	Simultaneous laterality judgment and precise lower-limb execution

Two Steps	ATT (sustained, divided), EF (strategic planning), LAT, VSP	Independent bilateral lower-limb dissociation to separate targets; asymmetric weight transfer	Maximum motor-cognitive integration; dissociated bilateral lower-limb control
Inside the Rubik Cube	ATT, EF (motor planning, timing), LAT, VSP	Simultaneous upper- and lower-limb coordination; four-limb integration; high balance	Maximum four-limb independent coordination under executive planning load
Follow the Trail	MEM (immediate spatial memory), ATT, EF (planning), VSP	Full-body spatial navigation reproducing a memorized path; sequential stepping	Spatial memory recall driving whole-body motor execution
Numerical Sequencing (displacement)	MEM (WM under locomotor load), EF, ATT, VSP	Full-body locomotion to numerical targets in correct sequence	Classic locomotor-cognitive dual-task: gait under WM load

S3b. Comparison with existing cognitive-motor intervention approaches

Table S3b distinguishes intervention approaches by two independent dimensions that are often conflated. Development purpose differentiates commercial off-the-shelf (COTS) exergames originally designed for entertainment or general fitness and later repurposed for rehabilitation from purpose-built serious exergames designed around predefined therapeutic aims. Immersion level describes the interface and may range from non-immersive conventional-screen systems to fully immersive head-mounted displays in either development category. The version of Kinesthesia evaluated in the 2017 study is therefore classified as a purpose-built, non-immersive therapeutic exergaming platform comprising 13 serious exergames.

The comparison is descriptive and uses explicit text rather than summary symbols because each literature category is heterogeneous. Entries describe typical design characteristics and should not be interpreted as universal properties or as evidence that one category is more effective than another.

Dimension	Sequential dual-task protocols	COTS exergames repurposed for rehabilitation	Purpose-built therapeutic exergames	Kinesthesia
Game-based interaction	Not required	Yes	Yes	Yes; 13-game study suite
Original development purpose	Cognitive-motor training protocol	Entertainment or general fitness	Predefined health or rehabilitation aims	Sensory-motor cognitive rehabilitation
Movement controls gameplay	Not necessarily	Yes	Yes	Yes; full-body motion capture
Cognitive-motor integration	May be simultaneous or sequential; components can remain separable	Depends on the selected commercial game and therapeutic adaptation	Built into the therapeutic task to varying degrees	Structurally integrated; interruption of either component prevents task completion
Clinical targeting and progression	Protocol- and therapist-dependent	Added after commercial development; game options may constrain tailoring	Specified during design; often configurable for the target population	Five cognitive-motor categories; bidirectional adaptation within and across sessions
Immersion level	Not applicable unless digitally mediated	Independent of purpose; may be non-, semi-, or fully immersive	Independent of purpose; may be non-, semi-, or fully immersive	Non-immersive; conventional screen without a head-mounted display
Performance monitoring	Protocol-dependent	Game- and platform-dependent	Platform-specific	Participant-level session database
Evidence interpretation	Heterogeneous programmes	Heterogeneous products and adaptations	Platform- and population-specific evidence	Uncontrolled feasibility pilot in participants spanning cognitively healthy to MCI; no comparative validation
Cost and home-delivery evidence	Variable and programme-specific	Procurement may be simpler, but cost-effectiveness is not assured	Often incompletely reported and platform-specific	Portable and accessible hardware architecture designed for clinical, community, home-based, and low-resource deployment; total implementation cost, cost-effectiveness, staff burden, and real-world scaling

Dimension	Sequential dual-task protocols	COTS exergames repurposed for rehabilitation	Purpose-built therapeutic exergames	Kinesthesia
				outcomes were not evaluated in this pilot

Representative evidence: sequential cognitive-motor training is characterised by Wollesen and Voelcker-Rehage (2014) and Karssemeijer et al. (2017); recreational and serious exergames are distinguished by Manser et al. (2025); COTS and purpose-built rehabilitation exergames are contrasted by Cieřlik (2025); and immersive versus non-immersive systems are compared by Huang (2020), Qin et al. (2025), and Hu et al. (2026). Full references are provided in the main manuscript.

Cost and implementation note: portability, accessibility, and deployability are intentional, implemented properties of the Kinesthesia architecture: the system uses a standard laptop, conventional display or television, compact depth sensor, modular software, and no head-mounted device. The present pilot did not measure total implementation cost, software maintenance, staff time, training requirements, cost-effectiveness, home delivery, or scaling outcomes. Continued use of the discontinued Kinect for Xbox 360 is a deliberate cost-conscious engineering decision rather than a consequence of limited technical capacity. A developer price check in September 2026 found replacement Kinect 360 units on Amazon, eBay, Mercado Libre, and other online marketplaces for under US\$50, whereas technically suitable Intel RealSense depth-camera alternatives were listed from approximately US\$400, with many configurations priced substantially higher. This dated descriptive snapshot—not a formal market analysis—documents the principal reason for retaining Kinect 360 and the non-immersive architecture. In more than ten years of platform use, none of the Kinect units used by the team has failed in service; three-dimensional skeletal tracking and the mature SDK satisfy all present game requirements. Greater precision and sampling rates offered by newer cameras exceed those requirements without a corresponding functional advantage for the current games. Long-term secondary-market availability cannot be guaranteed. Hardware-specific communication is confined to one sensor-interface module, allowing migration to a newer depth camera after adaptation and validation without redesigning the therapeutic game suite.

Supplementary Figures S4–S7. Spearman correlation heatmaps: game learning vs clinical outcomes

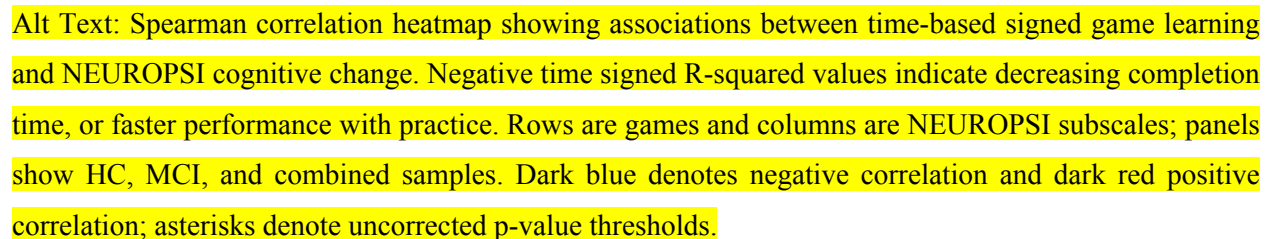
Figures S4–S7 present subgroup-level Spearman correlation heatmaps between within-platform learning and pre-to-post clinical change ($\Delta = \text{Post} - \text{Pre}$). Learning is represented as signed R^2 ($sR^2 = \text{sign of the within-participant regression slope} \times R^2$). Positive score sR^2 denotes increasing scores; negative time sR^2 denotes decreasing completion time and therefore faster performance with practice. Analyses are presented separately for HC, MCI, and the combined sample. Game-specific learning-sample n is shown in the row labels; outcome-specific pairwise n may be smaller where clinical data were missing. Only games with at

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least five participants showing a statistically significant learning slope ($p<0.05$) are included. Colour encodes Spearman ρ : blue = negative and red = positive; non-significant cells are white. Asterisks denote uncorrected associations (* $p<0.05$, ** $p<0.01$). The heatmaps are descriptive exploratory visualisations; BH-significant associations are identified in Table S1 and Figure 3.

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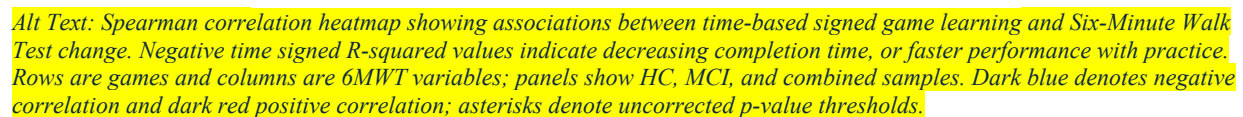
Supplementary Figure S4. Spearman ρ : signed game R^2 (time – speed learning) vs Δ NEUROPSY. Subgroup analyses (HC, MCI) and combined; $n \geq 5$, $p < 0.05$. Non-significant cells are white.



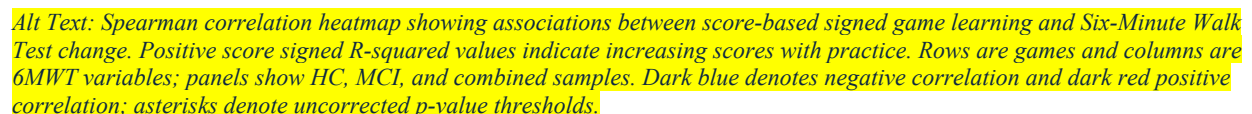
Alt Text: Spearman correlation heatmap showing associations between score-based signed game learning and NEUROPSI cognitive change. Positive score signed R-squared values indicate increasing scores with practice. Rows are games and columns are NEUROPSI subscales; panels show HC, MCI, and combined samples. Dark blue denotes negative correlation and dark red positive correlation; asterisks denote uncorrected p-value thresholds.



Supplementary Figure S6. Spearman ρ : signed game R^2 (time—speed learning) vs Δ 6MWI



Supplementary Figure S7. Spearman ρ : signed game R^2 (points—score learning) vs Δ 6MWT



Supplementary Appendix A. Investigator-developed post-intervention satisfaction questionnaire

Instrument status and administration. Satisfaction was assessed after the intervention with an investigator-developed Spanish-language questionnaire. It was not a previously published or psychometrically validated scale. The questionnaire collected numeric ratings for individual games or game groupings and for 12 equipment/service items, followed by questions about continued use and suggested improvements. Ratings were summarised descriptively; no psychometric total score or cut-off was interpreted. The item content below is an English translation of the archived Spanish form.

Response-format note. For the game ratings, the source form displayed columns numbered 1–5 with the following Spanish descriptors, reproduced verbatim: 1, “No muy satisfecho” (not very satisfied); 2, “Más o menos satisfecho” (more or less satisfied); 3, “Nada satisfecho” (not at all satisfied); 4, “satisfecho” (satisfied); and 5, “Muy satisfecho” (very satisfied). Thus, the archived form contains an apparent non-monotonic wording error among labels 1–3. The equipment and service sections instructed participants to enter a number from 1 to 5 but did not repeat separate verbal anchors. The present report therefore treats these responses as descriptive numeric satisfaction ratings and does not imply psychometric validation.

Questionnaire section	Item content (English translation)
Background fields	Name; date; gender; age; height; pathology or injury; current or previous sport and hours per week; education level; relationship with or experience using new technologies.
Game or game-group ratings (1–5)	1. Large Squares (Cuadros Grandes) 2. Small Squares (Cuadros Chicos) 3. Stars (Estrellas) 4. Skiing (Esquí) 5. Mathematics (Matemáticas) 6. Foot Games (Juegos de Pies) 7. Arm Games (Juegos de Brazos)
Equipment items (1–5)	How satisfied are you with: 1. The size of the screen? 2. The safety of the games? 3. The ease of performing the therapy? 4. The comfort of the games? 5. The number of videogames? 6. The types of activities in the videogames? 7. The usefulness of the games—do you think they are helpful in your life?
Service items (1–5)	1. What is your overall level of satisfaction with the technique? 2. What is your level of satisfaction with the session duration? 3. What is your overall level of satisfaction with the way you were treated? 4. What is your level of satisfaction with the spaces where you received the therapy? 5. What is your level of satisfaction with the number of therapy sessions received?
Additional questions	1. Would you like to continue attending therapy using this technique? 2. Would you like to acquire the equipment to use this technique at home? 3. Would you like to rent the equipment to use this technique at home? 4. What improvements would you make to the therapy (videogames, space, time, treatment)? 5. Which game did you like most? Which did you like least? Please provide suggestions for improvement.

Source-form clarification. The archived form states that the survey contains 12 questions; this corresponds to the seven equipment items and five service items. The seven game-specific ratings and additional follow-up questions also appear on the form.